

笔记前言：

本笔记的内容是去掉步骤的概述后，视频的所有内容。

本猴觉得，自己的步骤概述写的太啰嗦，大家自己做笔记时，

应该每个人都有自己的最舒服最简练的写法，所以没给大家写。

再是本猴觉得，不给大家写这个概述的话，大家会记忆的更深，

掌握的更好！

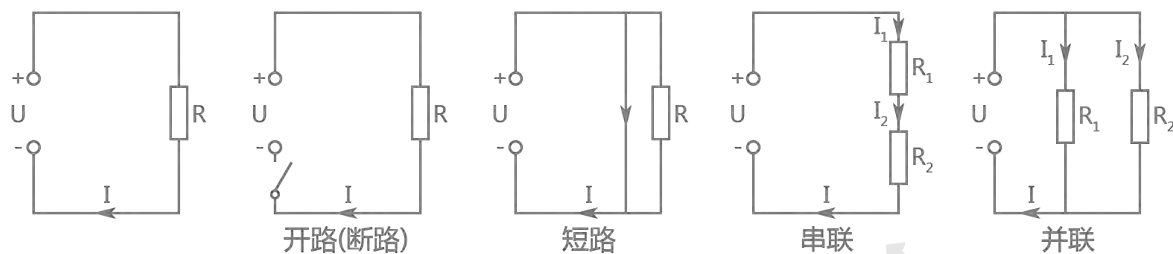
所以老铁！一定要过呀！不要辜负本猴的心意！~~~

【祝逢考必过，心想事成~~~~】

【一定能过！！！！！！】

电路第零课

一、串并联，短路与断路



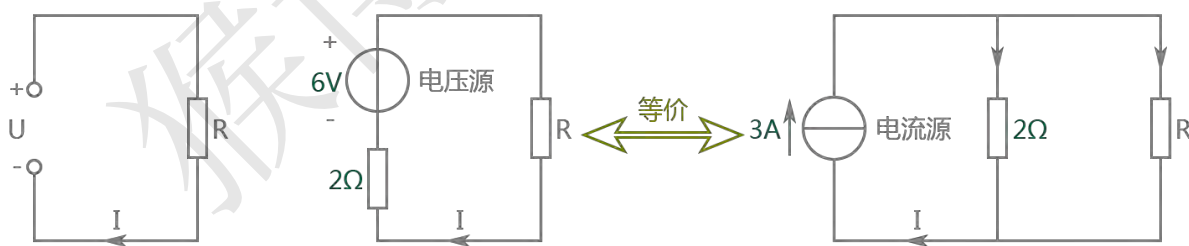
欧姆定律: $\frac{U}{I}=R$

串联: $I=I_1=I_2$, $R=R_1+R_2$, $U=U_1+U_2$

并联: $I=I_1+I_2$, $R=\frac{R_1 \cdot R_2}{R_1+R_2}$, $U=U_1=U_2$

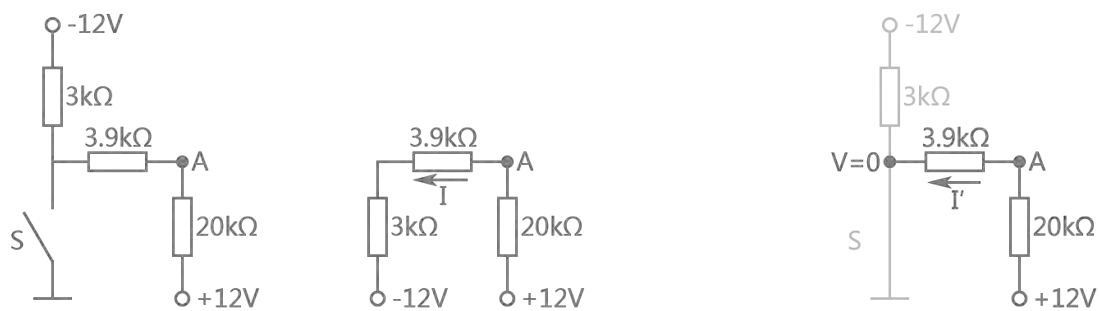
$$R=R_1 // R_2$$

二、电源



三、电位、电位差

例 1: 在下图中, 求 S 断开和闭合两种情况下 A 点的电位。



① S 断开时

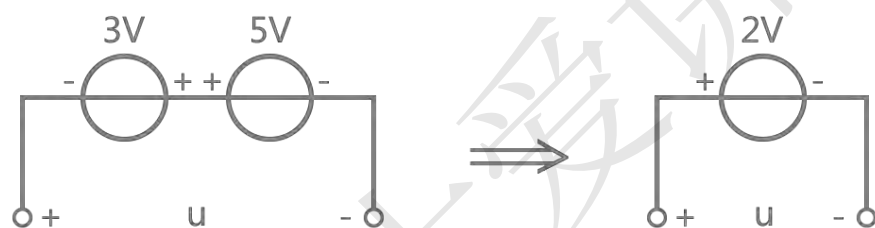
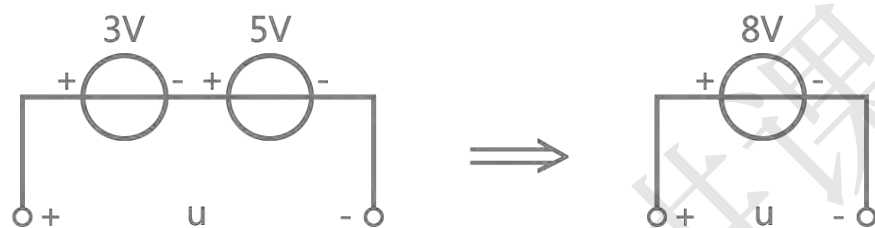
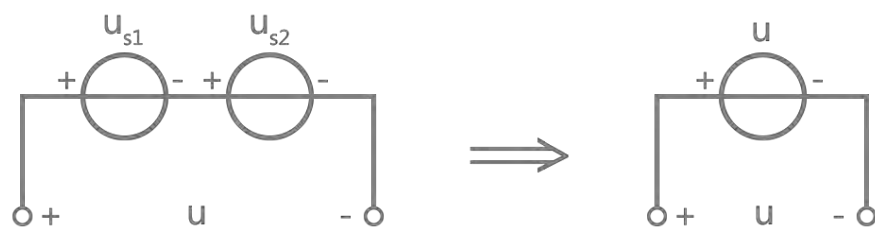
$$I = \frac{12 - (-12)}{20 + 3.9 + 3} = \frac{12 - V_A}{20} \Rightarrow V_A = -5.8V$$

② S 闭合时

$$I' = \frac{12 - 0}{20 + 3.9} = \frac{12 - V_A}{20} \Rightarrow V_A = 2V$$

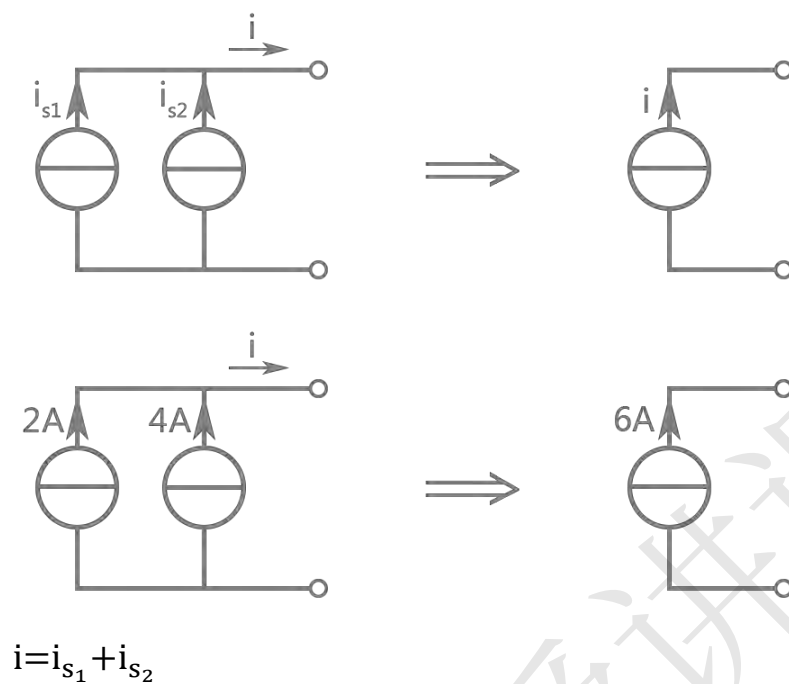
电路第一课

一、电压源串联

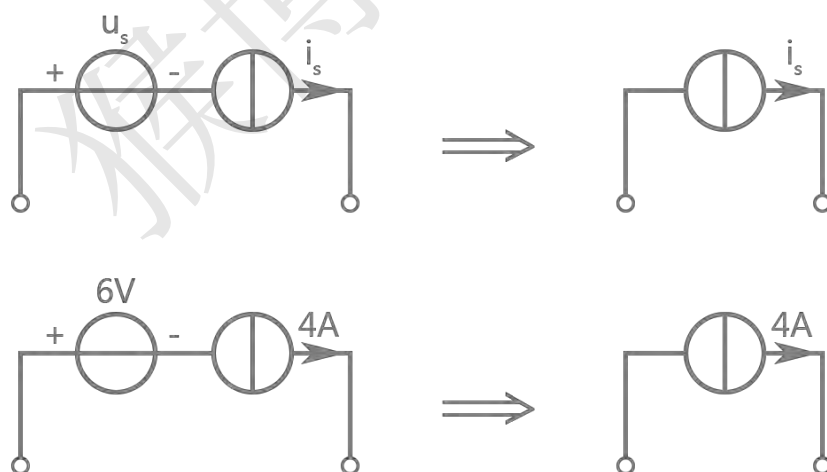


$$u = u_{s1} + u_{s2}$$

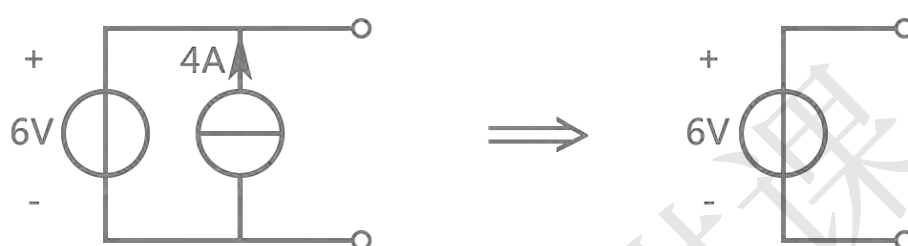
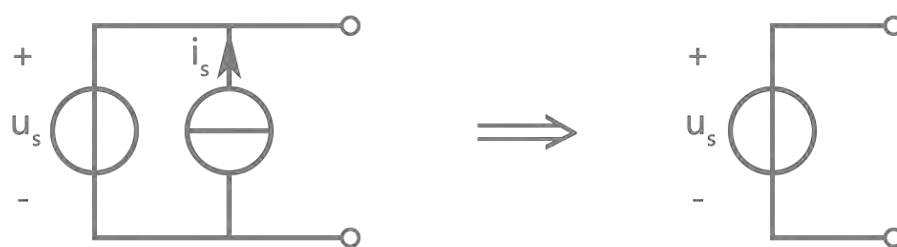
二、电流源并联



三、电压源和电流源串联



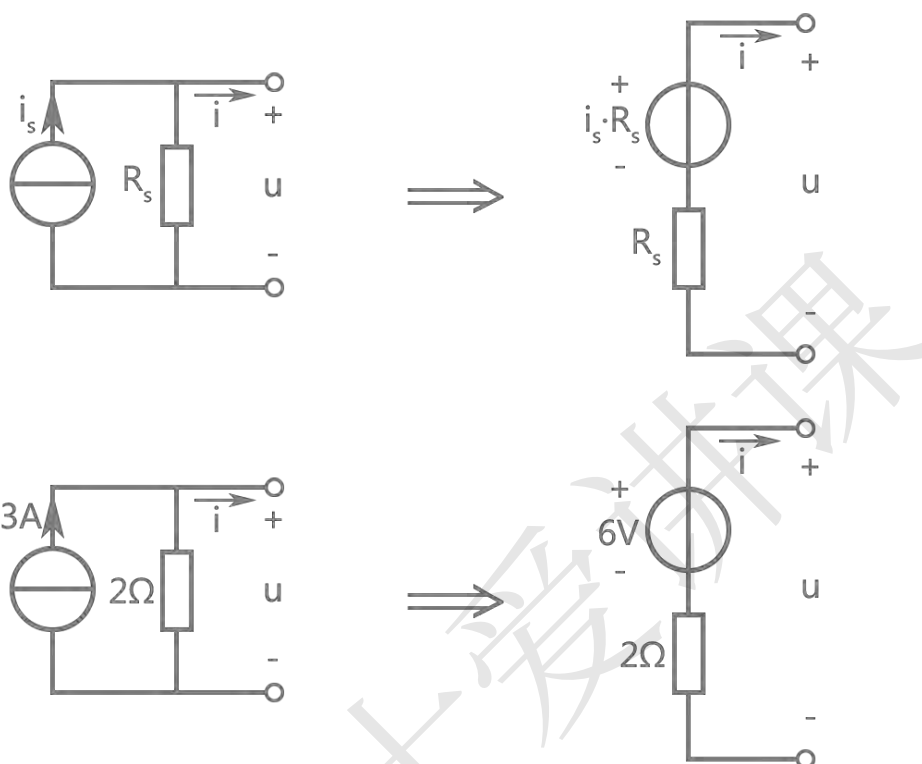
四、电压源和电流源并联



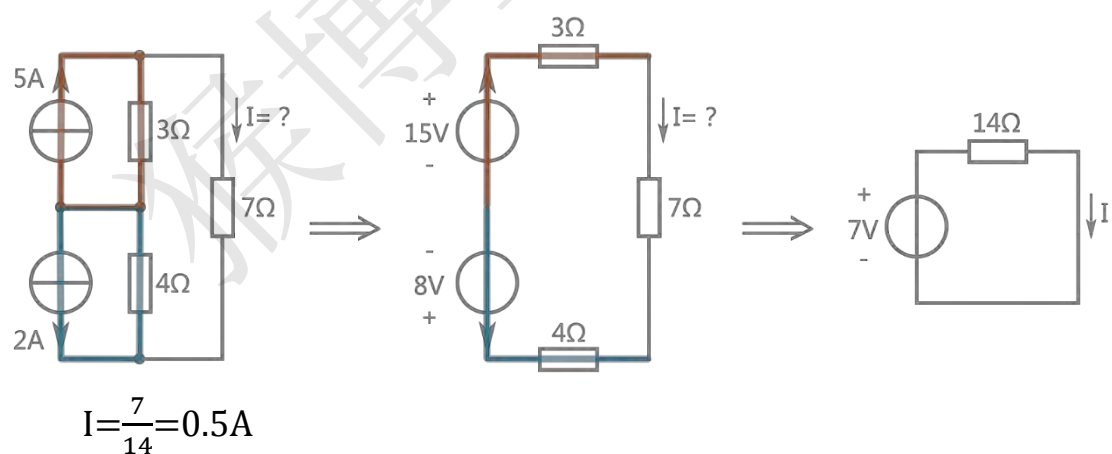
五、电压源转化为电流源



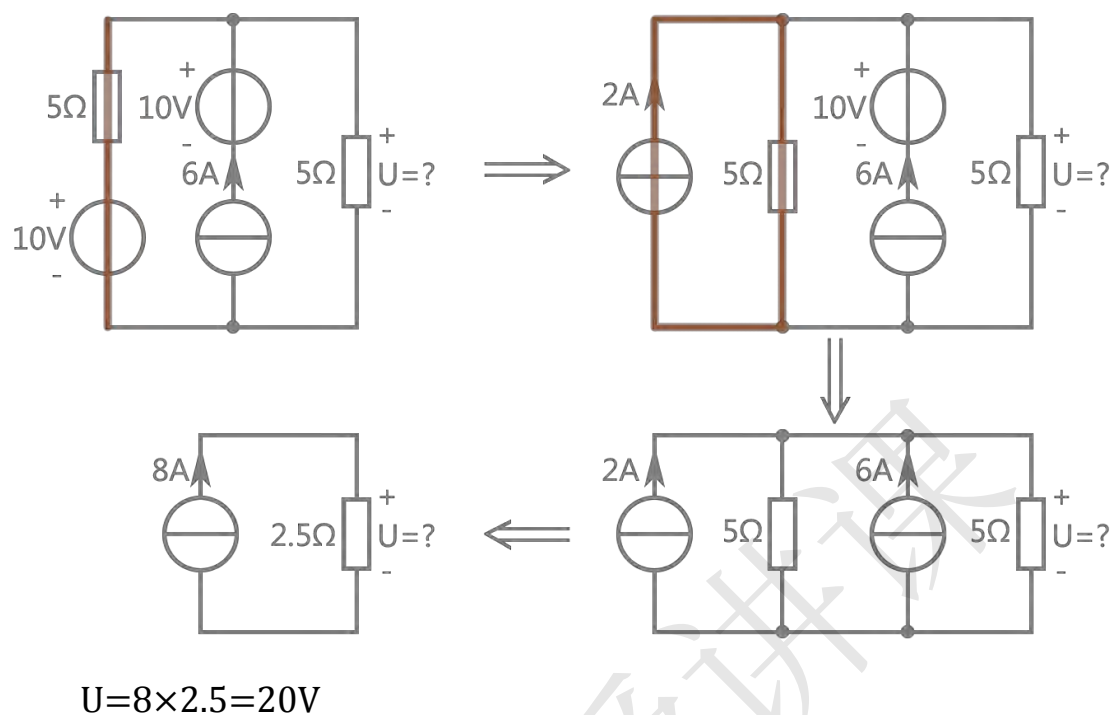
六、电流源转化成电压源



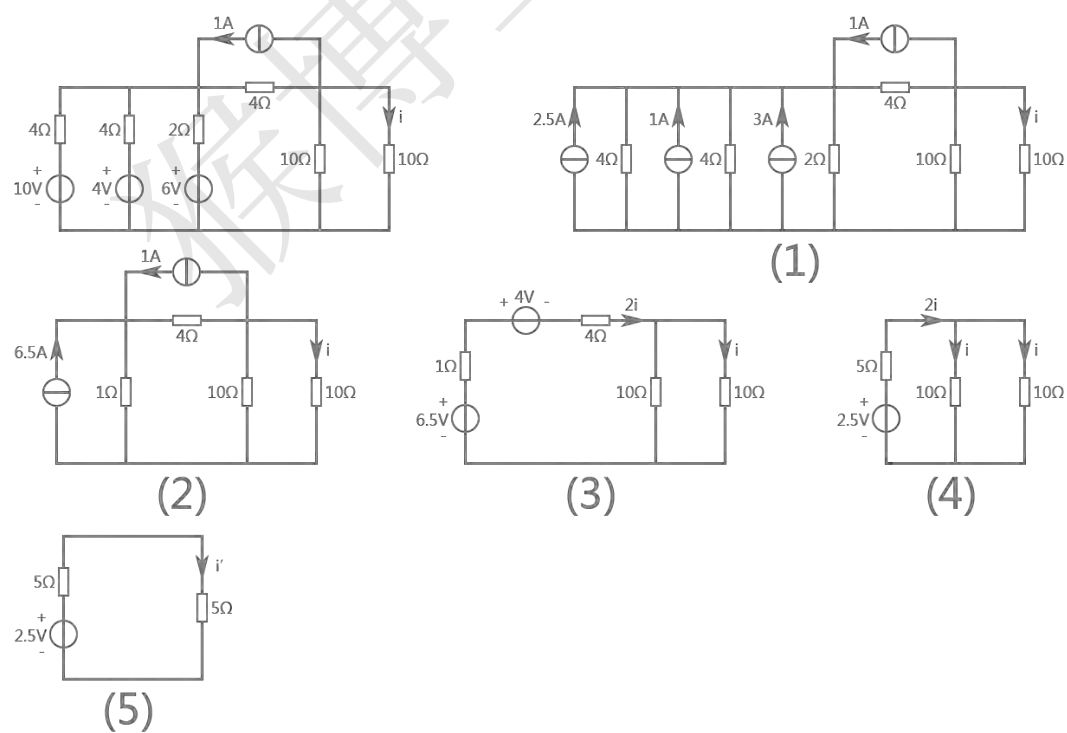
例 1：利用电源转换简化电路计算



例 2：利用电源转换简化电路计算



例 3：利用电源的等效变换，求图示电路中的电流 i

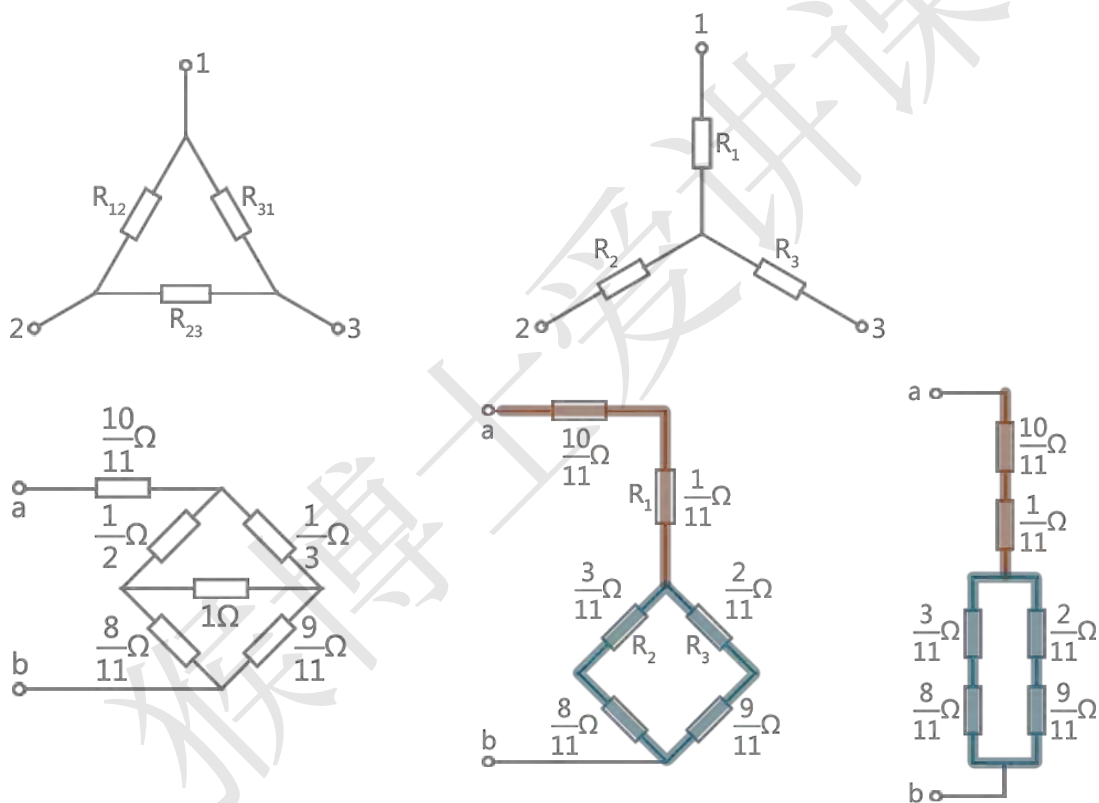


$$i' = \frac{2.5}{5+5} = 0.25\text{A}$$

$$u = i' \cdot 5 = 1.25\text{V}$$

$$i = 1.25 \div 10 = 0.125\text{A}$$

七、 Δ -Y 等效变换



Δ 变Y: Y形电阻 = $\frac{\Delta\text{形相邻电阻乘积}}{\Delta\text{形电阻之和}}$

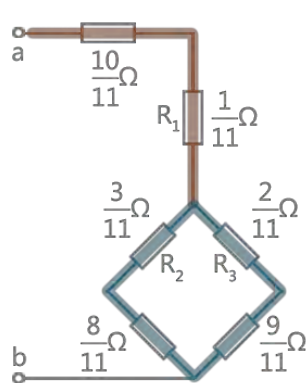
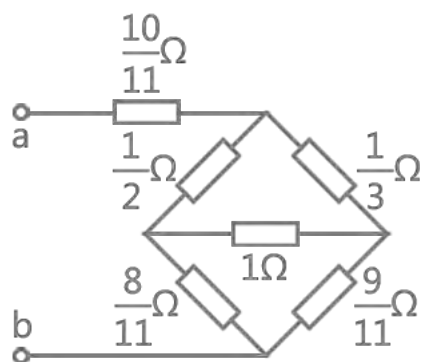
$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}} \quad R_2 = \frac{R_{12} \cdot R_{23}}{R_{12} + R_{23} + R_{31}} \quad R_3 = \frac{R_{23} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

Y变 Δ : Δ 形电阻 = $\frac{\text{Y形电阻两两乘积之和}}{\text{Y形不相邻电阻}}$

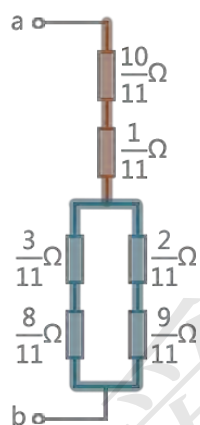
$$R_{12} = \frac{R_1 \cdot R_2 + R_2 \cdot R_3 + R_1 \cdot R_3}{R_3} \quad R_{23} = \frac{R_1 \cdot R_2 + R_2 \cdot R_3 + R_1 \cdot R_3}{R_1} \quad R_{31} = \frac{R_1 \cdot R_2 + R_2 \cdot R_3 + R_1 \cdot R_3}{R_2}$$

若三个电阻相等, 则 $R_{\Delta} = 3R_Y$

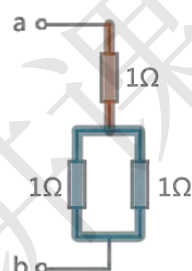
例 1：利用 Y-△等效变换求图中 a,b 端的等效电阻



(1)



(2)



(3)

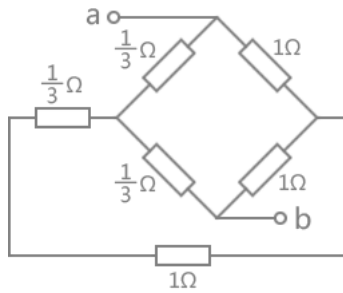
$$R_1 = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2} + \frac{1}{3} + 1} = \frac{1}{11} \Omega$$

$$R_2 = \frac{\frac{1}{2} \times 1}{\frac{1}{2} + \frac{1}{3} + 1} = \frac{3}{11} \Omega$$

$$R_3 = \frac{\frac{1}{3} \times 1}{\frac{1}{2} + \frac{1}{3} + 1} = \frac{2}{11} \Omega$$

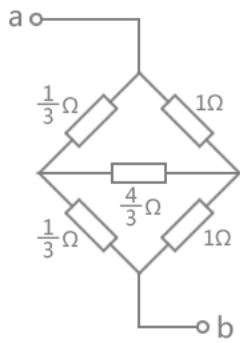
$$R_{ab} = 1 + \frac{1 \times 1}{1 + 1} = 1.5 \Omega$$

例 2：求图示电路中的等效电阻 R_{ab}

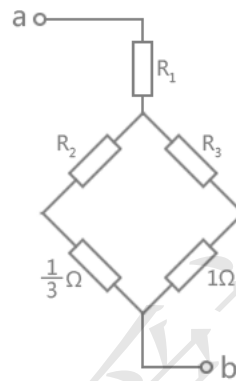


(1)

$$\frac{1}{3}\Omega + 1\Omega = \frac{4}{3}\Omega$$



(2)

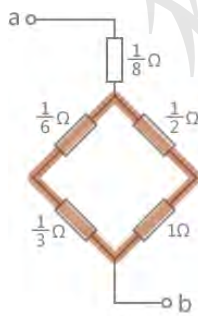


(3)

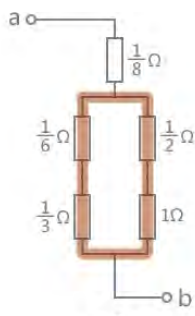
$$R_1 = \frac{\frac{1}{3} \times 1}{\frac{1}{3} + 1 + \frac{1}{3}} = \frac{1}{8}\Omega$$

$$R_2 = \frac{\frac{1}{3} \times \frac{4}{3}}{\frac{1}{3} + 1 + \frac{1}{3}} = \frac{1}{6}\Omega$$

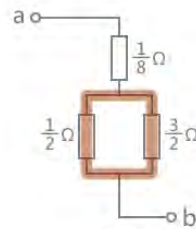
$$R_3 = \frac{1 \times \frac{4}{3}}{\frac{1}{3} + 1 + \frac{1}{3}} = \frac{1}{2}\Omega$$



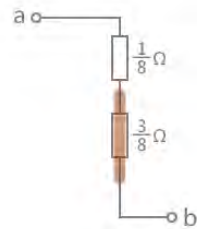
(4)



(5)



(6)



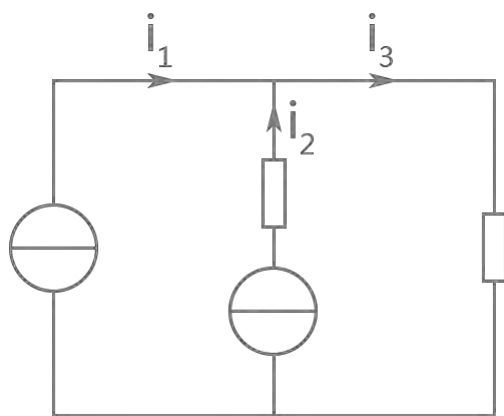
(7)

$$R_{ab} = \frac{1}{8}\Omega + \frac{3}{8}\Omega = \frac{1}{2}\Omega$$

电路第二课

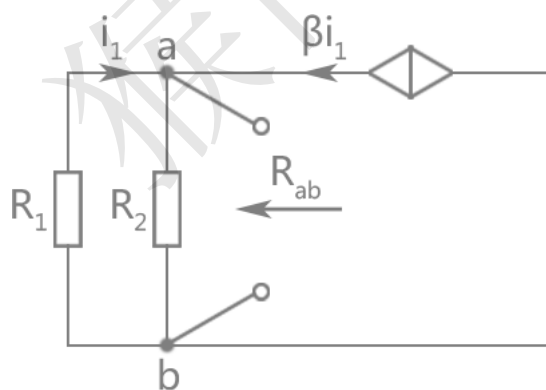
一、基尔霍夫电流定律

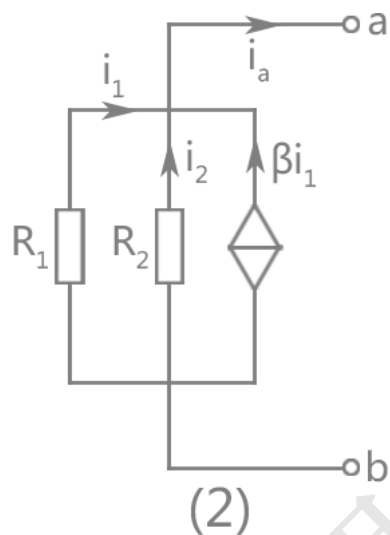
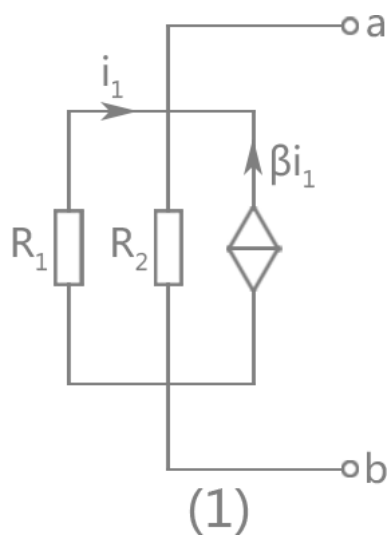
基尔霍夫电流定律：任一结点(交汇点)上流出电流之和等于流入电流之和。



$$i_1 + i_2 = i_3$$

例 1：求图示电路中的输入电阻 R_{ab}





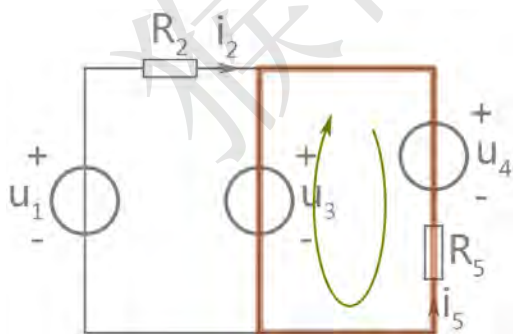
$$i_2 = \frac{u}{R_2} = \frac{i_1 \cdot R_1}{R_2}$$

$$i_a = i_1 + i_2 + \beta i_1 = \left(1 + \beta + \frac{R_1}{R_2}\right) i_1$$

$$R_{ab} = \frac{u_{ab}}{i_a} = \frac{i_1 \cdot R_1}{\left(1 + \beta + \frac{R_1}{R_2}\right) i_1} = \frac{R_1}{1 + \beta + \frac{R_1}{R_2}}$$

二、基尔霍夫电压定律

基尔霍夫电压定律：任一回路上 U 与 IR 之和为 0

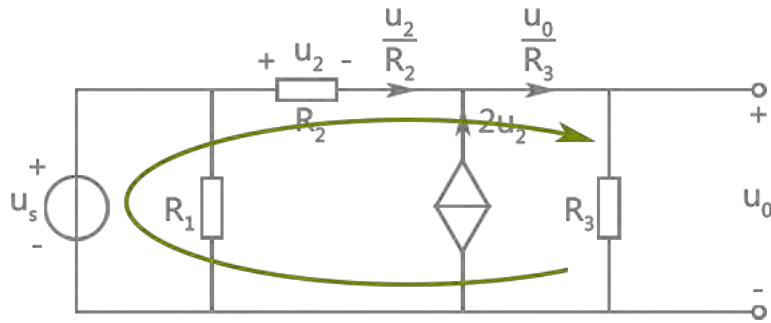


$$-u_1 + i_2 R_2 + u_4 - i_5 R_5 = 0$$

$$-u_1 + i_2 R_2 + u_3 = 0$$

$$-u_3 + u_4 - i_5 R_5 = 0$$

例 1: 已知 $R_1=R_2=R_3=1\Omega$, 求电路中的电压比 $\frac{u_0}{u_s}$



$$\because \frac{u_2}{R_2} + 2u_2 = \frac{u_0}{R_3}$$

$$\therefore u_0 = 3u_2$$

$$\because -u_s + u_2 + u_0 = 0$$

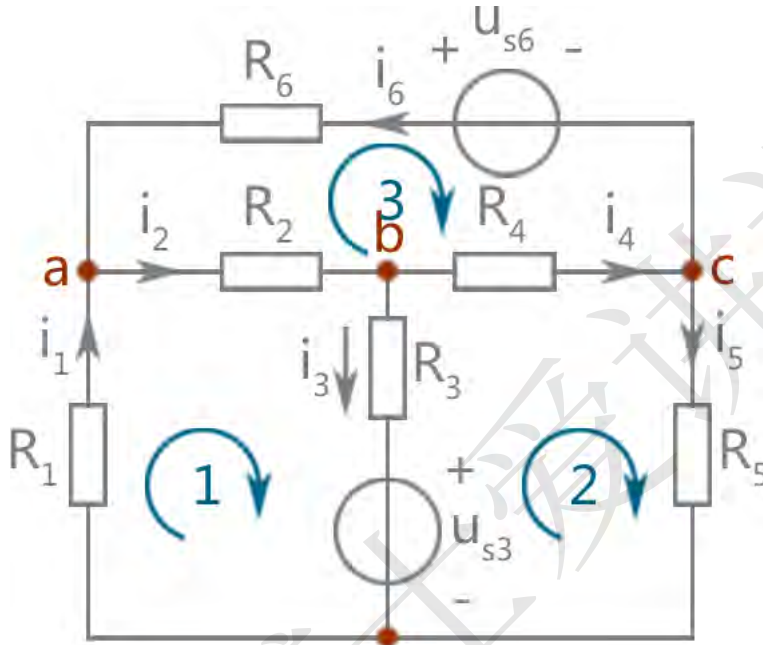
$$\therefore u_s = u_2 + u_0 = u_2 + 3u_2 = 4u_2$$

$$\text{则 } \frac{u_0}{u_s} = \frac{3u_2}{4u_2} = \frac{3}{4}$$

电路第三课

一、支路电流法

例 1: 如图所示, 电路中 $R_1=R_2=10\Omega$, $R_3=4\Omega$, $R_4=R_5=8\Omega$, $R_6=2\Omega$, $u_{s3}=20V$, $u_{s6}=40V$, 用支路电流法求解电流 i_5



$$\begin{cases} \text{结点 a: } i_1 + i_6 = i_2 \\ \text{结点 b: } i_2 = i_3 + i_4 \\ \text{结点 c: } i_4 = i_5 + i_6 \end{cases}$$

支路数: 6 $6 - 4 + 1 = 3$

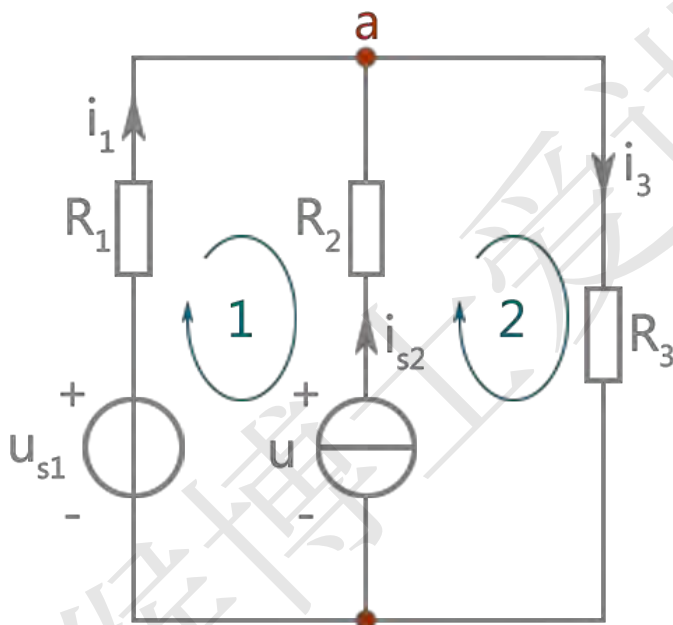
回路 1: $i_1 R_1 + i_2 R_2 + i_3 R_3 + u_{s3} = 0$

回路 2: $-i_3 R_3 + i_4 R_4 + i_5 R_5 - u_{s3} = 0$

回路 3: $-i_2 R_2 - i_6 R_6 - i_4 R_4 + u_{s6} = 0$

$$\begin{cases} i_1 + i_6 = i_2 \\ i_2 = i_3 + i_4 \\ i_4 = i_5 + i_6 \\ 10i_1 + 10i_2 + 4i_3 = -20 \\ -4i_3 + 8i_4 + 8i_5 = 20 \\ -10i_2 - 8i_4 - 2i_6 = -40 \end{cases} \Rightarrow i_5 = -0.956A$$

例 2：如图所示，电路中 $R_1=7\Omega$ ， $R_2=11\Omega$ ， $R_3=7\Omega$ ， $u_{s1}=70V$ ， $i_{s2}=6A$ ，用支路电流法求解电流 i_3



结点数：2 结点 a: $i_1 + i_{s2} = i_3$

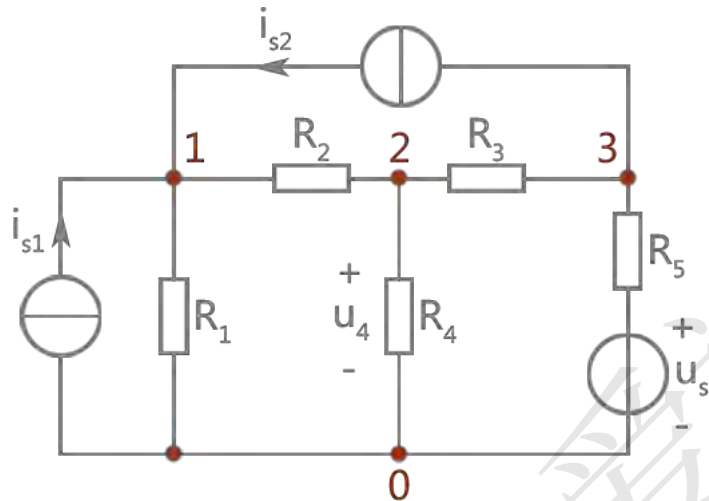
支路数：3 $3 - 2 + 1 = 2$

$$\begin{cases} \text{回路 1: } i_1 R_1 - i_{s2} R_2 + u - u_{s1} = 0 \\ \text{回路 2: } i_{s2} R_2 + i_3 R_3 - u = 0 \end{cases}$$

$$\begin{cases} i_1 + 6 = i_3 \\ 7i_1 - 6 \times 11 + u - 70 = 0 \\ 6 \times 11 + 7i_3 - u = 0 \end{cases} \Rightarrow i_3 = 8A$$

二、结点电压法

例 1: 如图所示, 电路中 $R_1=1\Omega$, $R_2=2\Omega$, $R_3=3\Omega$, $R_4=6\Omega$, $R_5=2\Omega$, $i_{s1}=4A$, $i_{s2}=2A$, $u_s=36V$, 用结点电压法求解电压 u_4



$$G_{11} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{3}{2}$$

$$G_{22} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} = 1$$

$$G_{33} = \frac{1}{R_3} + \frac{1}{R_5} = \frac{5}{6}$$

$$G_{12} = G_{21} = -\frac{1}{R_2} = -\frac{1}{2}$$

$$G_{23} = G_{32} = -\frac{1}{R_3} = -\frac{1}{3}$$

$$G_{13} = G_{31} = 0$$

$$i_{s11} = i_{s1} + i_{s2} = 2 + 4 = 6A$$

$$i_{s22} = 0$$

$$i_{s33} = -i_{s2} + \frac{u_s}{R_5} = 16A$$

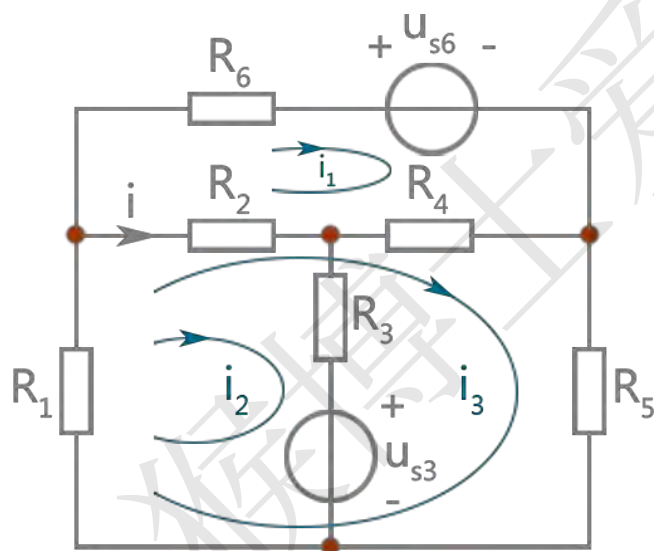
$$\begin{cases} G_{11}u_{n1} + G_{12}u_{n2} + G_{13}u_{n3} = i_{s11} \\ G_{21}u_{n1} + G_{22}u_{n2} + G_{23}u_{n3} = i_{s22} \\ G_{31}u_{n1} + G_{32}u_{n2} + G_{33}u_{n3} = i_{s33} \end{cases}$$

$$\begin{cases} \frac{3}{2}u_{n1} - \frac{1}{2}u_{n2} = 6 \\ -\frac{1}{2}u_{n1} + u_{n2} - \frac{1}{3}u_{n3} = 0 \\ -\frac{1}{3}u_{n2} + \frac{5}{6}u_{n3} = 16 \end{cases} \Rightarrow \begin{cases} u_{n1} = 8V \\ u_{n2} = 12V \\ u_{n3} = 24V \end{cases}$$

$$u_4 = 12V$$

三、回路(网孔)电流法

例 1: 如图所示, 电路中 $R_1=R_2=10\Omega$, $R_3=4\Omega$, $R_4=R_5=8\Omega$, $R_6=2\Omega$, $u_{s6}=-33V$, $u_{s3}=-12.2V$, 用回路电流法求解电流 i



支路数:6 结点数: 4 $6-4+1=3$

$$R_{11}=R_2+R_4+R_6=10+8+2=20\Omega$$

$$R_{22}=R_1+R_2+R_3=10+10+4=24\Omega$$

$$R_{33}=R_1+R_2+R_4+R_5=10+10+8+8=36\Omega$$

$$R_{12}=R_{21}=-R_2=-10\Omega$$

$$R_{13}=R_{31}=-R_2 - R_4=-18\Omega$$

$$R_{23}=R_{32}=R_2+R_1=10+10=20\Omega$$

$$u_{s_{11}}=-u_{s_6}=33V$$

$$u_{s_{22}}=-u_{s_3}=12.2V$$

$$u_{s_{33}}=0$$

$$\begin{cases} R_{11}i_1 + R_{12}i_2 + R_{13}i_3 = u_{s_{11}} \\ R_{21}i_1 + R_{22}i_2 + R_{23}i_3 = u_{s_{22}} \\ R_{31}i_1 + R_{32}i_2 + R_{33}i_3 = u_{s_{33}} \end{cases}$$

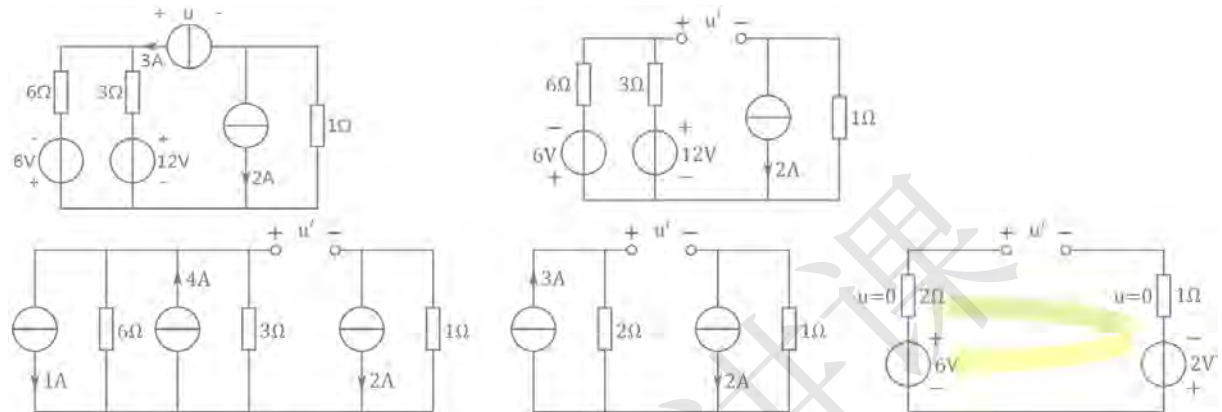
$$\begin{cases} 20i_1 - 10i_2 - 18i_3 = 33 \\ -10i_1 + 24i_2 + 20i_3 = 12.2 \\ -18i_1 + 20i_2 + 36i_3 = 0 \end{cases} \Rightarrow \begin{cases} i_1 = 3A \\ i_2 = 0.95A \\ i_3 = 0.97A \end{cases}$$

$$i = i_2 + i_3 - i_1 = -1.08A$$

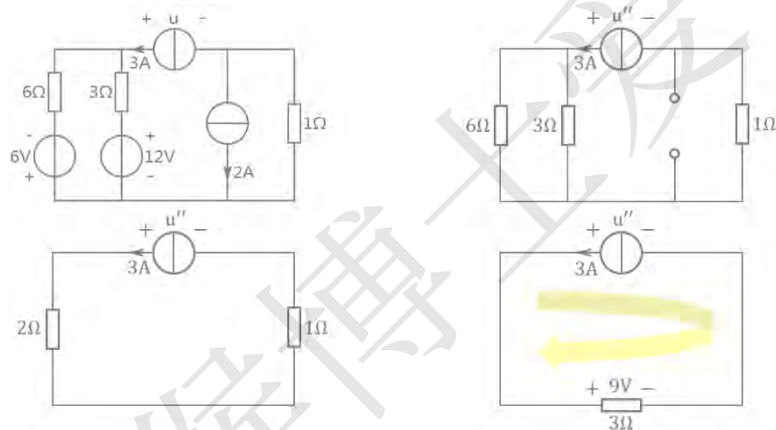
电路第四课

一、叠加定理

例 1：如图所示电路，用叠加定理求电压 u



$$-6V + u' - 2V = 0 \Rightarrow u' = 8V$$

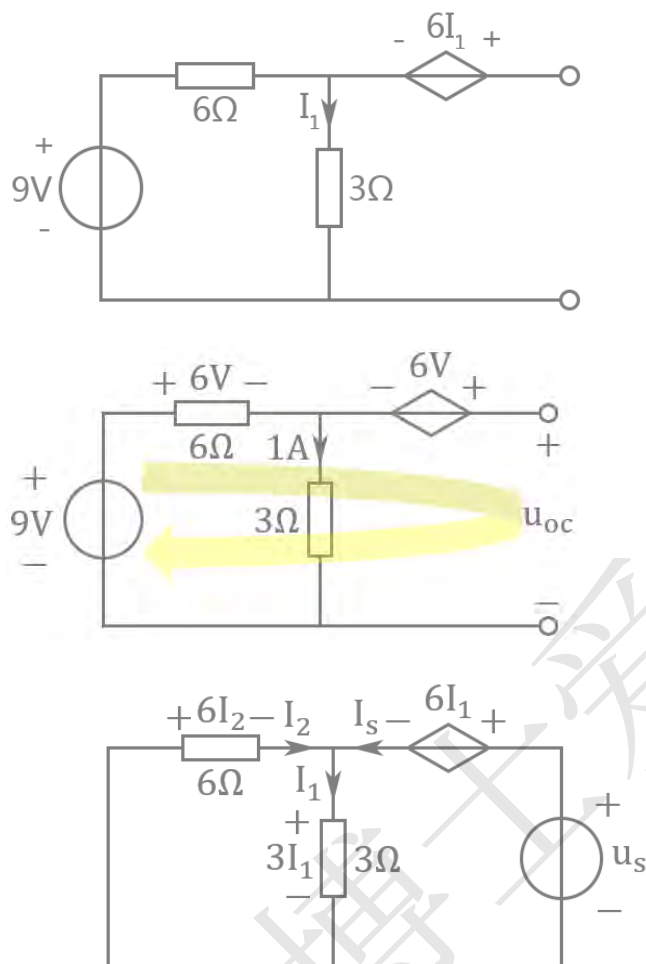


$$-9V + u'' = 0 \Rightarrow u'' = 9V$$

$$u = u' + u'' = 8 + 9 = 17V$$

二、戴维宁等效电路

例 1：求该端口的戴维宁等效电路。



$$I_1 = \frac{9V}{6\Omega + 3\Omega} = 1A$$

$$u_{6\Omega} = 1A \times 6\Omega = 6V$$

$$-9V + 6V - 6V + u_{oc} = 0 \Rightarrow u_{oc} = 9V$$

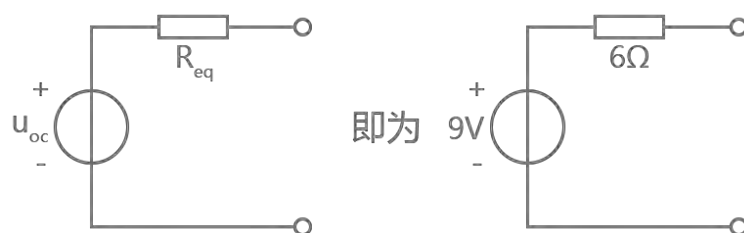
$$6I_2 + 3I_1 = 0 \Rightarrow I_1 = -2I_2$$

$$I_2 + I_s = I_1 = -2I_2 \Rightarrow I_s = -3I_2$$

$$6I_2 - 6I_1 + u_s = 0 \Rightarrow u_s = 6I_1 - 6I_2 = -18I_2$$

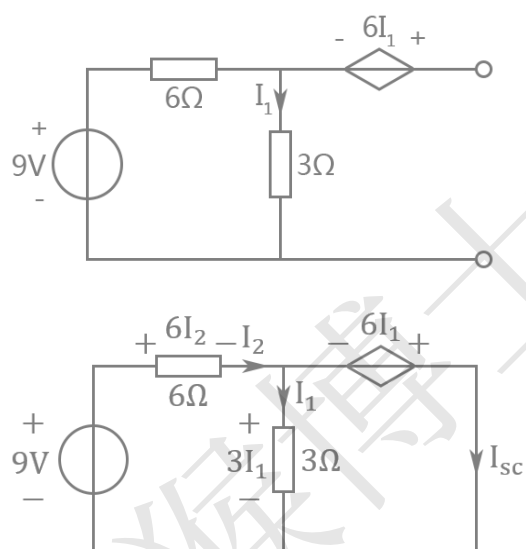
$$\therefore R_{eq} = \frac{u_s}{I_s} = \frac{-18I_2}{-3I_2} = 6\Omega$$

戴维宁等效电路为：



三、诺顿等效电路

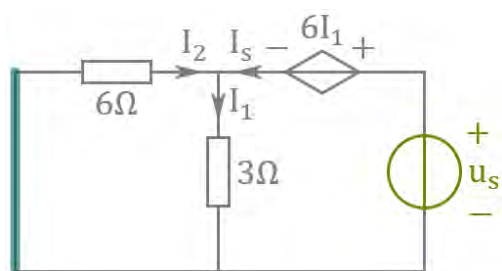
例 1：求该端口的诺顿等效电路。



$$-3I_1 - 6I_1 = 0 \Rightarrow I_1 = 0$$

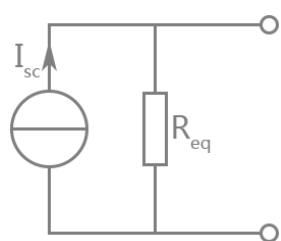
$$-9 + 6I_2 + 0 = 0 \Rightarrow I_2 = 1.5\text{A}$$

$$1.5\text{A} = 0 + I_{sc} \Rightarrow I_{sc} = 1.5\text{A} - 0 = 1.5\text{A}$$

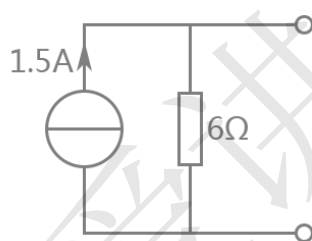


$$\therefore R_{eq} = \frac{u_s}{I_s} = 6\Omega$$

诺顿等效电路为：

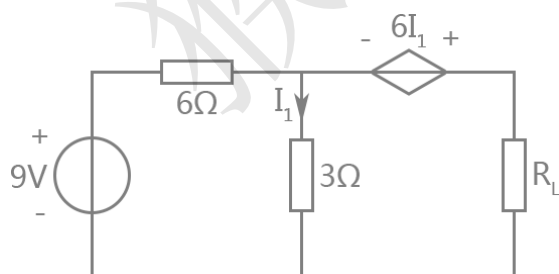


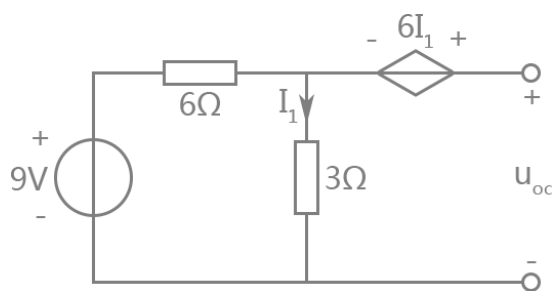
即为



四、求某电阻阻值为何值时可吸收最大功率

例 1：图示电路中， R_L 取何值才能获得最大功率？并求此最大功率。

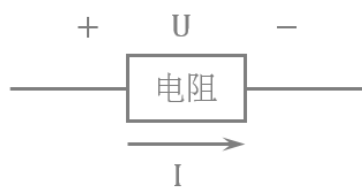




$$u_{oc}=9V, R_{eq}=6\Omega$$

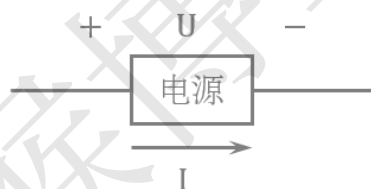
$$\text{当 } R_L=6\Omega \text{ 时功率最大, } P_{\max}=\frac{9^2}{4 \times 6}=3.375W$$

五、吸收/释放功率



$$P=UI$$

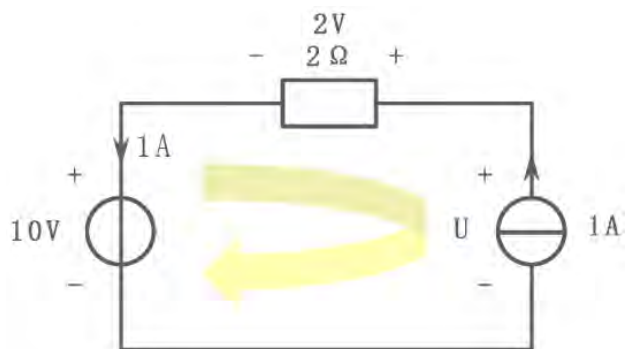
(恒定吸收功率)



$$P=UI$$

(I从高电压流向低电压则吸收功率，反之则释放功率)

例 1：试判断下图中的电阻、电压源、电流源各吸收或释放了多少功率。



$$P_{\text{电阻}} = 2V \cdot 1A = 2w \text{ (吸收)}$$

$$P_{\text{电压源}} = 10V \cdot 1A = 10w \text{ (吸收)}$$

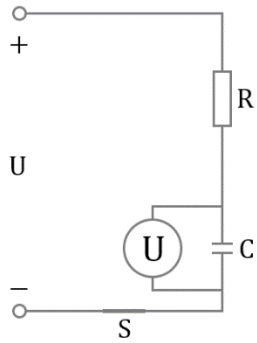
$$-10V - 2V + U = 0 \rightarrow U = 12V$$

$$P_{\text{电流源}} = 12V \cdot 1A = 12w \text{ (释放)}$$

电路第五课

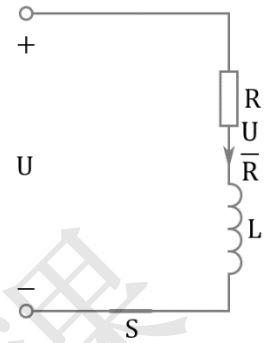
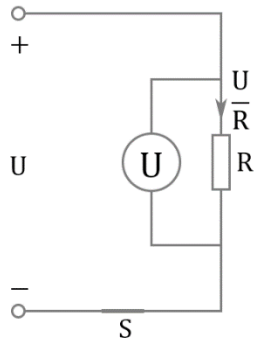
$$u_C = u_C(\infty) + [u_C(0_+) - u_C(\infty)]e^{-\frac{t-t_0}{\tau}} \quad (\tau = RC) \quad i_L = i_L(\infty) + [i_L(0_+) - i_L(\infty)]e^{-\frac{t-t_0}{\tau}} \quad (\tau = \frac{L}{R})$$

$$i_C = C \frac{du_C}{dt} \quad u_L = L \frac{di_L}{dt}$$



阻止电压突变

$$U_C = ?t$$

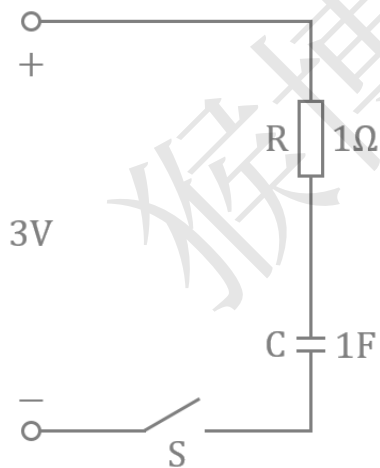


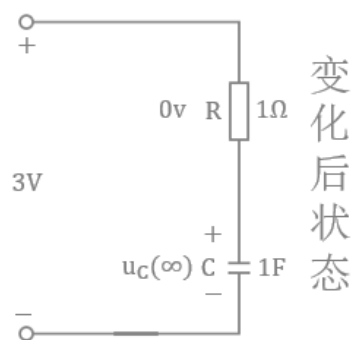
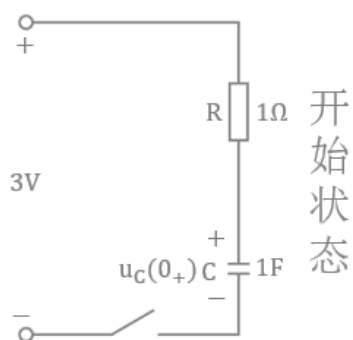
阻止电流突变

$$I_L = ?t$$

一、RC 电路的响应

例 1: 如图, 开关 S 在 3s 时闭合, 试求开关闭合后, 电容两端的电压、电流同时间的关系

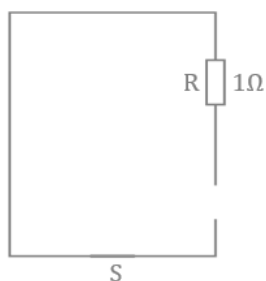




$$u_C(0_+) = 0$$

$$-3 + u_R + u_C(\infty) = 0 \Rightarrow u_C(\infty) = 3V$$

$$t_0 = 3s$$

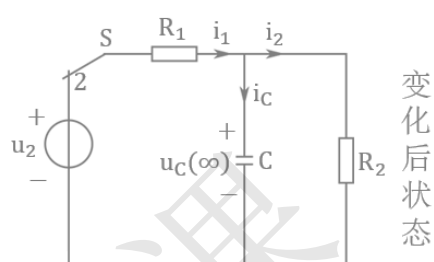
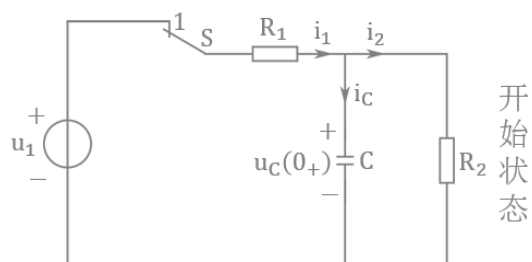
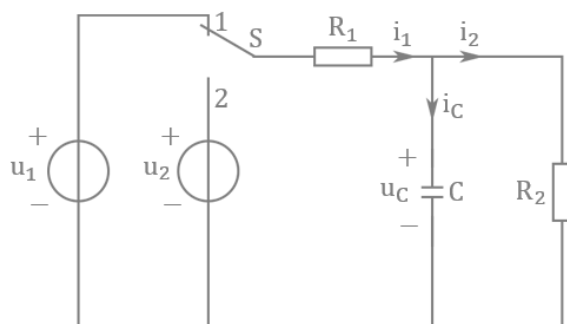


$$R = 1\Omega \Rightarrow \tau = RC = 1s$$

$$u_C = 3 + (0 - 3)e^{-\frac{t-3}{1}} = 3 - 3e^{-t+3}V$$

$$i_C = 1 \times (3 - 3e^{-t+3})' = 3e^{-t+3} A \Rightarrow i_C(0_+) = 3e^{-3+3} = 3 A$$

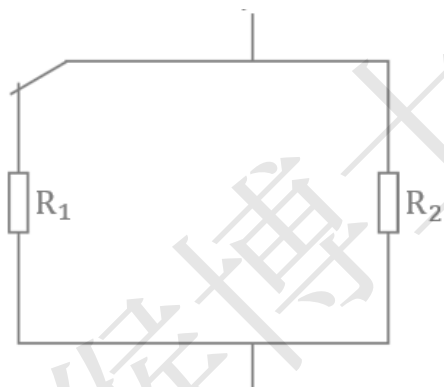
例 2：如图所示电路，开关处在 1 位置， $t=0$ 时，将开关合到 2 位置，求电容两端的电压 u_C 。(已知： $u_1=3V$ ， $u_2=5V$ ， $R_1=1k\Omega$ ， $R_2=2k\Omega$ ， $C=3\mu F$)



$$\therefore u_C(0_+) = u_1 \cdot \frac{R_2}{R_1 + R_2} = 3 \times \frac{2}{1+2} = 2V$$

$$\therefore u_C(\infty) = u_2 \cdot \frac{R_2}{R_1 + R_2} = 5 \times \frac{2}{1+2} = \frac{10}{3}V$$

$$t_0 = 0$$

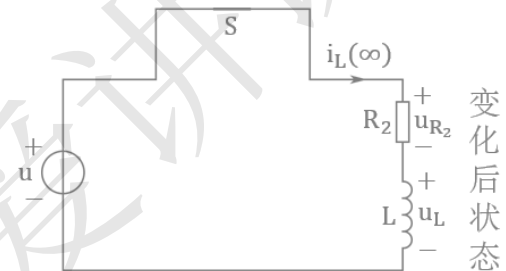
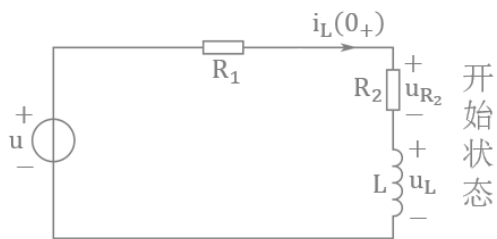
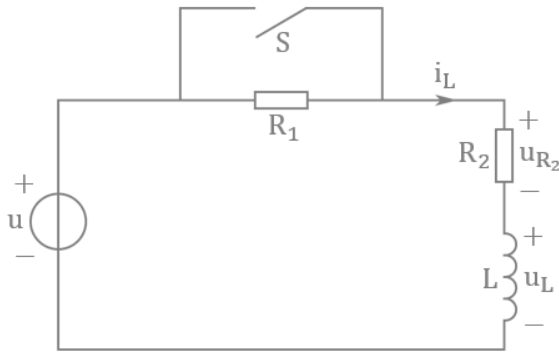


$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{1 \times 2}{1+2} = \frac{2}{3} \times 10^3 \Omega \Rightarrow \tau = RC = \frac{2}{3} \times 10^3 \times 3 \times 10^{-6} = 2 \times 10^{-3} s$$

$$u_C = \frac{10}{3} + (2 - \frac{10}{3})e^{-\frac{t-0}{2 \times 10^{-3}}} = \frac{10}{3} - \frac{4}{3}e^{-500t} V$$

二、RL 电路的响应

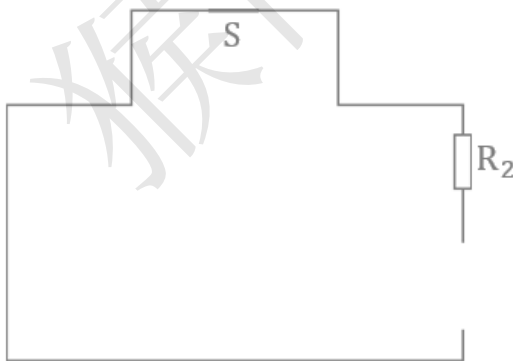
例 1：如图所示电路，开始开关是断开的， $t=0$ 时，开关闭合，求流过电感的电流。(已知： $u=220\text{V}$ ， $L=0.6\text{H}$ ， $R_1=8\Omega$ ， $R_2=12\Omega$)



$$i_L(0_+) = \frac{u}{R_1 + R_2} = \frac{220}{8 + 12} = 11\text{A}$$

$$i_L(\infty) = \frac{u}{R_2} = \frac{220}{12} = 18.3\text{A}$$

$$t_0 = 0$$



$$R = R_2 = 12\Omega$$

$$\therefore \tau = \frac{L}{R} = \frac{0.6}{12} = 0.05\text{s}$$

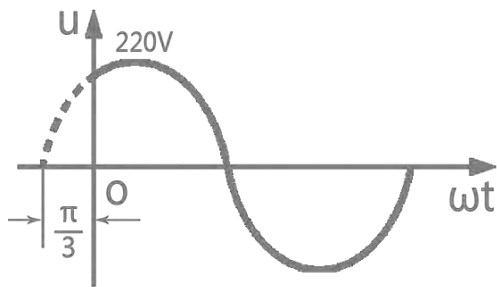
$$i_L = 18.3 + (11 - 18.3)e^{-\frac{t-0}{0.05}} = (18.3 - 7.3e^{-20t}) \text{ A}$$

开始	后来	名称
通电	不通电	零输入响应
不通电	通电	零状态响应
通电	通电	全响应

电路第六课

一、看图写正弦交流变量的表达式

例 1：下图正弦交流电压频率为 50Hz，试写出其表达式并计算出有效值。



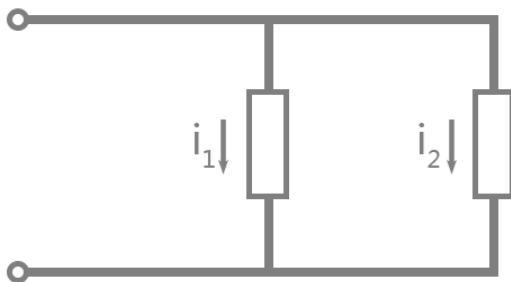
$$u = U_m \sin(\omega t + \varphi)$$

$$u = 220 \sin\left(100\pi t + \frac{\pi}{3}\right)$$

$$U = \frac{U_m}{\sqrt{2}} = \frac{220}{\sqrt{2}} \text{V}$$

二、交流变量的相加与乘除

例 1：图中 $i_1 = 30\sqrt{2}\sin(\pi t + 45^\circ)$ ， $i_2 = 40\sqrt{2}\sin(\pi t + 60^\circ)$ ，试求总电流 i 的表达式。

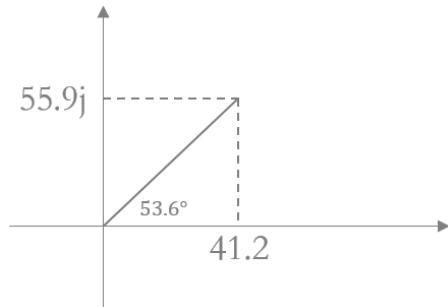


$$\dot{I}_1 = 30(\cos 45^\circ + j \cdot \sin 45^\circ) = 30\left(\frac{\sqrt{2}}{2} + j \cdot \frac{\sqrt{2}}{2}\right) = 15\sqrt{2} + 15\sqrt{2}j$$

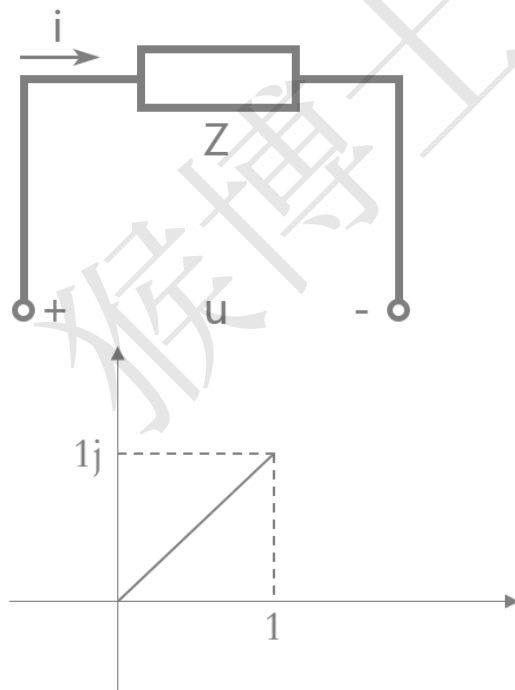
$$\dot{I}_2 = 40(\cos 60^\circ + j \cdot \sin 60^\circ) = 40\left(\frac{1}{2} + j \cdot \frac{\sqrt{3}}{2}\right) = 20 + 20\sqrt{3}j$$

$$\dot{I} = \dot{I}_1 + \dot{I}_2 = (15\sqrt{2} + 20) + j(15\sqrt{2} + 20\sqrt{3}) = 41.2 + 55.9j$$

$$i = \sqrt{2} \cdot \sqrt{41.2^2 + 55.9^2} \sin(\pi t + 53.6^\circ)$$



例 2：已知图中电压 $u = 5\sqrt{2}\sin(2t + 35^\circ)$ ， $Z=1+j$ ，试求电流 i 的表达式。



$$\dot{U}=5\angle 35^{\circ}$$

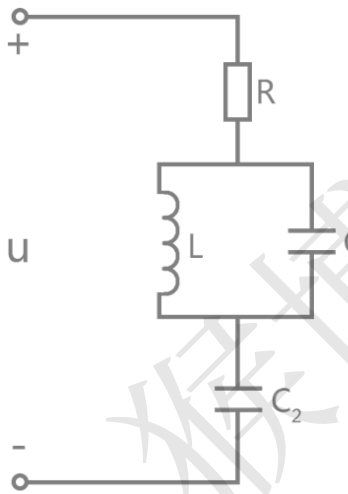
$$Z=\sqrt{2}\angle 45^{\circ}$$

$$\dot{i}=\frac{\dot{U}}{Z}=\frac{5}{\sqrt{2}}\angle 35^{\circ}-45^{\circ}=\frac{5}{\sqrt{2}}\angle -10^{\circ}$$

$$i=5\sin(\omega t-10^{\circ})=5\sin(2t-10^{\circ})$$

三、求电路的总阻抗

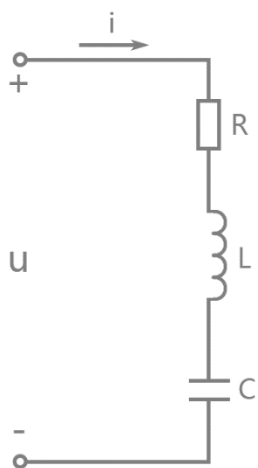
例 1: 已知 $u = 10\sqrt{2}\sin(10t + 30^{\circ})$, $R = 2\Omega$, $L = 100\text{mH}$, $C_1 = 0.2\text{F}$, $C_2 = 0.3\text{F}$, 试求电路中的总阻抗。



$$\begin{aligned} Z_{\text{总}} &= Z_R + Z_L // Z_{C_1} + Z_{C_2} \\ &= 2 + 0.1j\omega // \frac{-j}{0.2\omega} + \frac{-j}{0.3\omega} \\ &= 2 + \frac{0.1j\omega \times \frac{-j}{0.2\omega}}{0.1j\omega + \frac{-j}{0.2\omega}} + \frac{-j}{0.3\omega} \\ &= 2 + \frac{0.1j \times 10 \times \frac{-j}{0.2 \times 10}}{0.1j \times 10 + \frac{-j}{0.2 \times 10}} + \frac{-j}{0.3 \times 10} \\ &= 2 - \frac{4}{3}j \end{aligned}$$

四、画相量图

例 1: 已知 $u = 10\sqrt{2}\sin(10t + 30^\circ)$, $R = 2\Omega$, $L = 100\text{mH}$, $C = 0.2\text{F}$ 。试求电流 i 与各部分电压 u_R , u_C , u_L 的表达式。并画出他们的相量图。

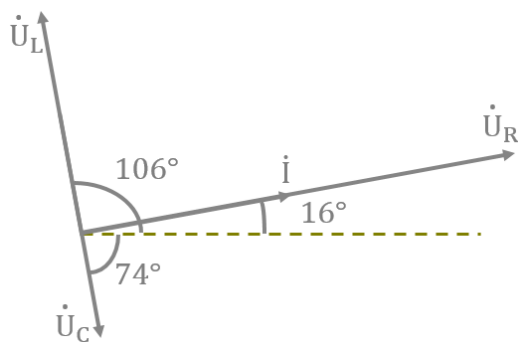


$$\dot{I} = 4.76 \angle 16^\circ$$

$$\dot{U}_R = 9.52 \angle 16^\circ$$

$$\dot{U}_L = 4.76 \angle 106^\circ$$

$$\dot{U}_C = 2.38 \angle -74^\circ$$



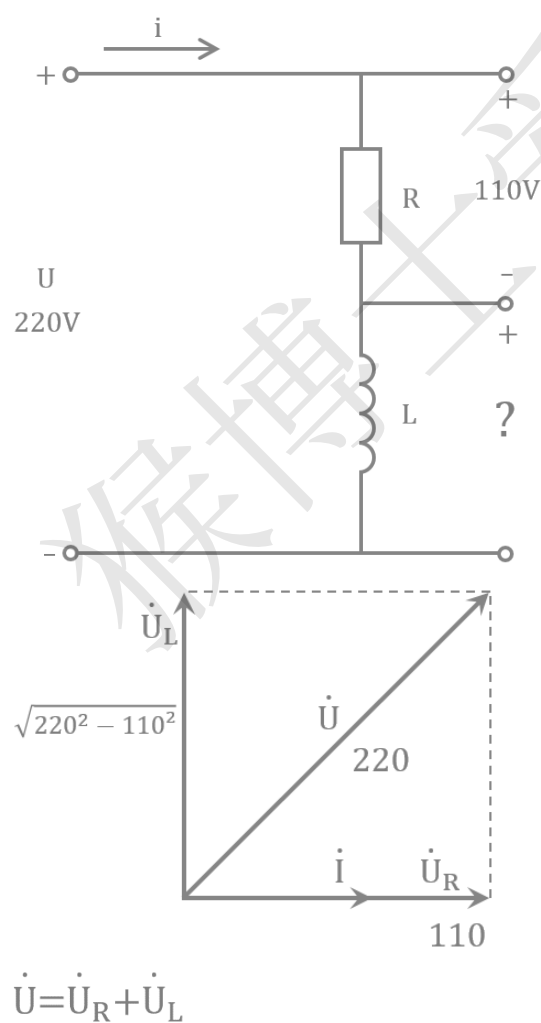
经过 R, U_R 与 I 同相

经过 L, U_L 比 I 超前 90°

经过 C, U_C 比 I 滞后 90°

五、利用相量图做题

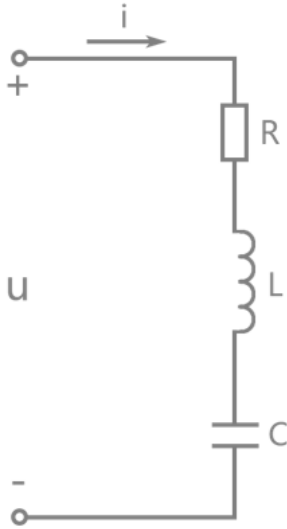
例 1: 已知 $U=220V$, 电阻 R 两端电压为 $110V$, 试求电感 L 两端电压的有效值。



电路第七课

一、求有功功率与无功功率

例 1: 已知 $u = 10\sqrt{2}\sin(10t + 30^\circ)$, $R = 2\Omega$, $L = 100\text{mH}$, $C = 0.2\text{F}$ 。试求各部分以及总电路的有功功率与无功功率。



$$Z_R = 2\angle 0^\circ \quad Z_L = 1\angle 90^\circ \quad Z_C = 0.5\angle -90^\circ \quad Z_{\text{总}} = 2.1\angle 14^\circ$$

$$\dot{U}_R = 9.52\angle 16^\circ \quad \dot{U}_L = 4.76\angle 106^\circ \quad \dot{U}_C = 2.38\angle -74^\circ \quad \dot{U} = 10\angle 30^\circ$$

$$\dot{I} = 4.76\angle 16^\circ$$

$$P_R = \frac{U_R^2}{|Z_R|} \cos\varphi_R = \frac{9.52^2}{2} \times \cos 0^\circ = 45.32 \text{ W}$$

$$Q_R = \frac{U_R^2}{|Z_R|} \sin\varphi_R = \frac{9.52^2}{2} \times \sin 0^\circ = 0 \text{ var}$$

$$P_L = \frac{U_L^2}{|Z_L|} \cos\varphi_L = \frac{4.76^2}{1} \times \cos 90^\circ = 0 \text{ W}$$

$$Q_L = \frac{U_L^2}{|Z_L|} \sin\varphi_L = \frac{4.76^2}{1} \times \sin 90^\circ = 22.66 \text{ var}$$

$$P_C = \frac{U_C^2}{|Z_C|} \cos\varphi_C = \frac{2.38^2}{0.5} \times \cos(-90^\circ) = 0 \text{ W}$$

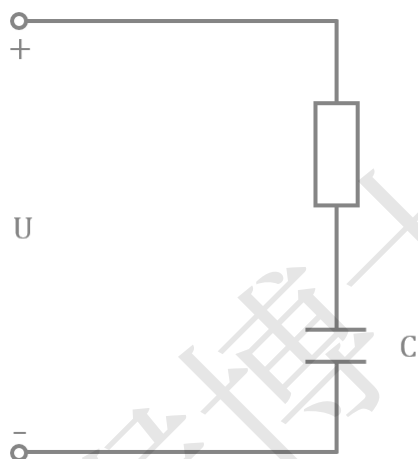
$$Q_C = \frac{U_C^2}{|Z_C|} \sin\varphi_C = \frac{2.38^2}{0.5} \times \sin(-90^\circ) = -11.33 \text{ var}$$

$$P_{\text{总}} = \frac{U_{\text{总}}^2}{|Z_{\text{总}}|} \cos \varphi_{\text{总}} = \frac{10^2}{2.1} \times \cos 14^\circ = 46.19 \text{ W}$$

$$Q_{\text{总}} = \frac{U_{\text{总}}^2}{|Z_{\text{总}}|} \sin \varphi_{\text{总}} = \frac{10^2}{2.1} \times \sin 14^\circ = 11.52 \text{ var}$$

二、利用功率因数求功率

例 1：电路功率因数 $\cos \varphi = 0.8$ ，有功功率 $P = 1\text{kW}$ ，求电容产生的无功功率。



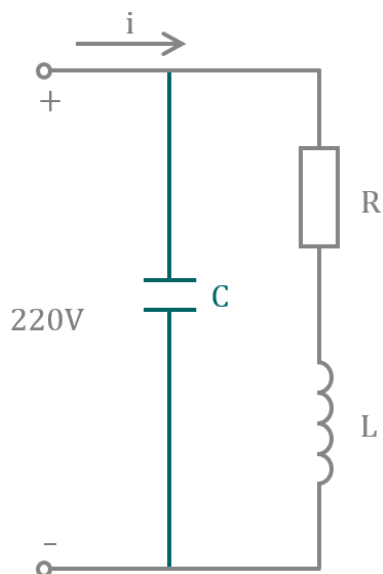
$$\because \cos \varphi = 0.8$$

$$\therefore \sin \varphi = \sqrt{1 - 0.8^2} = 0.6$$

$$\therefore \tan \varphi = \frac{\sin \varphi}{\cos \varphi} = 0.75 = \frac{Q_{\text{总}}}{P_{\text{总}}}$$

$$\therefore Q_{\text{总}} = 0.75 P_{\text{总}} = 0.75 \text{ kvar}$$

例 2：如图，电感性负载，其功率 $P=10\text{kW}$ ，功率因数 $\cos\varphi_1=0.6$ ，电源电压 220V ，电源频率 50Hz ，并联一个电容后，功率因数提高到 $\cos\varphi_2=0.95$ ，求电容 C 产生的无功功率 Q_C 。



并联前： $\cos\varphi_1=0.6 \Rightarrow \sin\varphi_1=\sqrt{1-0.6^2}=0.8$

$$\Rightarrow \tan\varphi_1=\frac{0.8}{0.6}=\frac{Q_L}{P}$$

$$\therefore Q_L=P\cdot\frac{0.8}{0.6}=13.33\text{kvar}$$

并联后： $\cos\varphi_2=0.95 \Rightarrow \sin\varphi_2=\sqrt{1-0.95^2}\approx 0.3$

$$\Rightarrow \tan\varphi_2=\frac{0.3}{0.95}=\frac{Q_{\text{总}}}{P_{\text{新}}}$$

$$\therefore P_{\text{新}}=P=10\text{kW}$$

$$\therefore Q_{\text{总}}=P\cdot\frac{0.3}{0.95}=3.29\text{kvar}$$

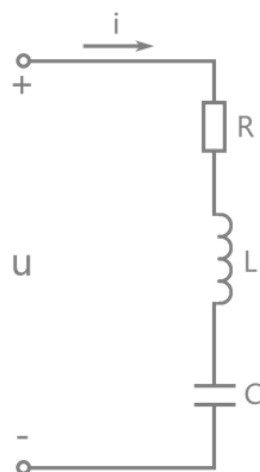
$$\therefore Q_{\text{总}}=Q_C+Q_L=Q_C+13.33=3.29$$

$$\therefore Q_C=-10\text{kvar}$$

三、求视在功率与复功率

例 1: 已知 $u = 10\sqrt{2}\sin(10t + 30^\circ)$, $R = 2\ \Omega$, $L = 100\text{mH}$, $C = 0.2\text{F}$ 。

试求各部分以及总电路的有功功率与无功功率。以及视在功率和复功率。



$$Z_R = 2 \angle 0^\circ$$

$$Z_L = 1 \angle 90^\circ$$

$$Z_C = 0.5 \angle -90^\circ$$

$$Z_{\text{总}} = 2.1 \angle 14^\circ$$

$$\dot{U}_R = 9.52 \angle 16^\circ$$

$$\dot{U}_L = 4.76 \angle 106^\circ$$

$$\dot{U}_C = 2.38 \angle -74^\circ$$

$$\dot{U} = 10 \angle 30^\circ$$

$$\dot{I} = 4.76 \angle 16^\circ$$

$$P_R = \frac{U_R^2}{|Z_R|} \cos \varphi_R = \frac{9.52^2}{2} \times \cos 0^\circ = 45.32\text{ W}$$

$$Q_R = \frac{U_R^2}{|Z_R|} \sin \varphi_R = \frac{9.52^2}{2} \times \sin 0^\circ = 0\text{ var}$$

$$P_L = \frac{U_L^2}{|Z_L|} \cos \varphi_L = \frac{4.76^2}{1} \times \cos 90^\circ = 0\text{ W}$$

$$Q_L = \frac{U_L^2}{|Z_L|} \sin \varphi_L = \frac{4.76^2}{1} \times \sin 90^\circ = 22.66\text{ var}$$

$$P_C = \frac{U_C^2}{|Z_C|} \cos \varphi_C = \frac{2.38^2}{0.5} \times \cos (-90^\circ) = 0\text{ W}$$

$$Q_C = \frac{U_C^2}{|Z_C|} \sin \varphi_C = \frac{2.38^2}{0.5} \times \sin (-90^\circ) = -11.33\text{ var}$$

$$P_{\text{总}} = \frac{U_{\text{总}}^2}{|Z_{\text{总}}|} \cos \varphi_{\text{总}} = \frac{10^2}{2.1} \times \cos 14^\circ = 46.19 \text{ W}$$

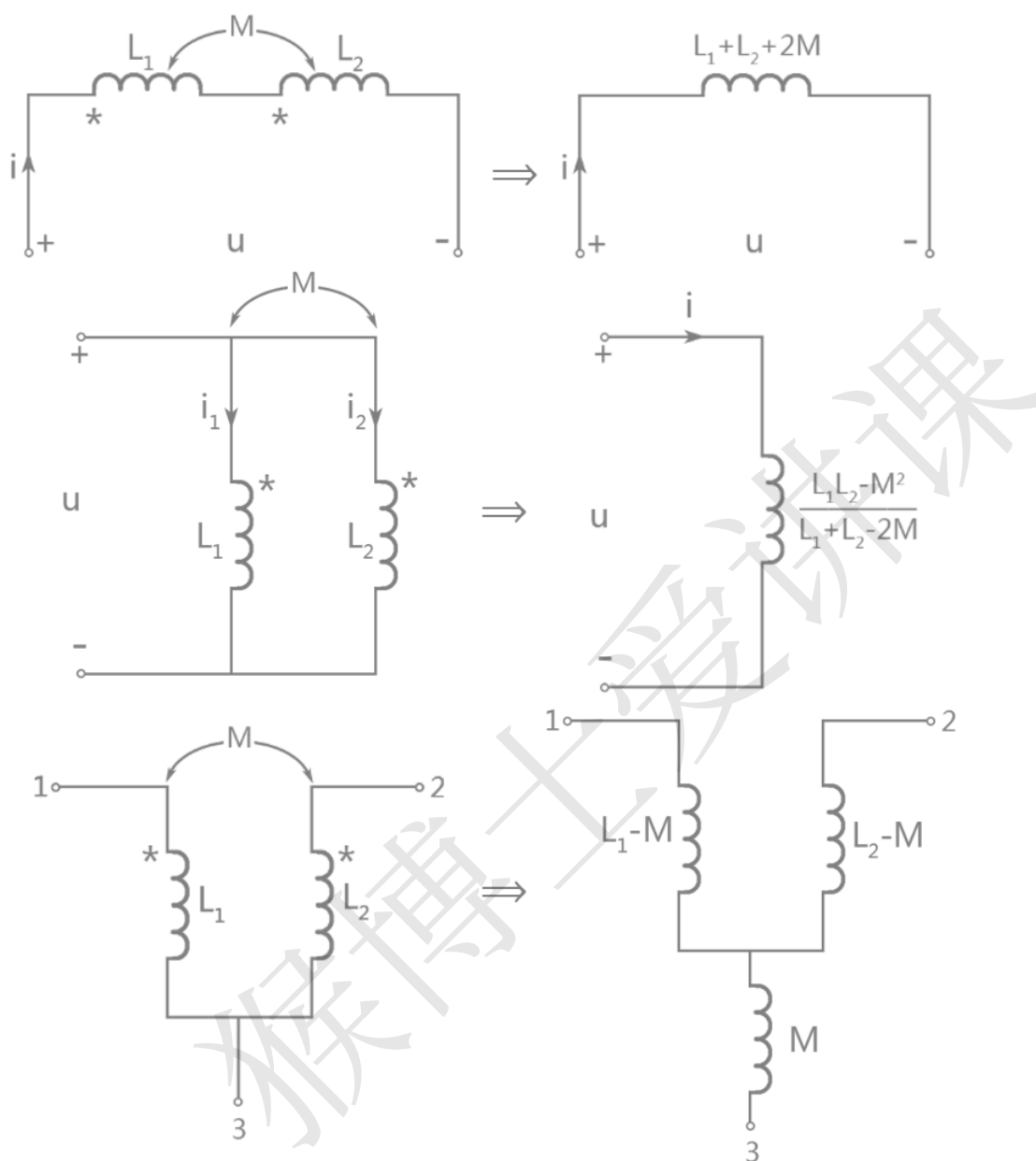
$$Q_{\text{总}} = \frac{U_{\text{总}}^2}{|Z_{\text{总}}|} \sin \varphi_{\text{总}} = \frac{10^2}{2.1} \times \sin 14^\circ = 11.52 \text{ var}$$

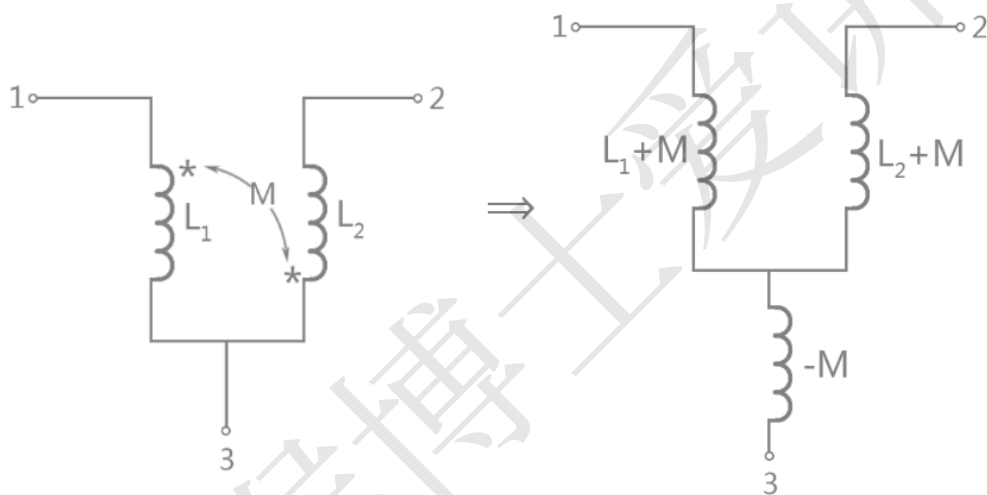
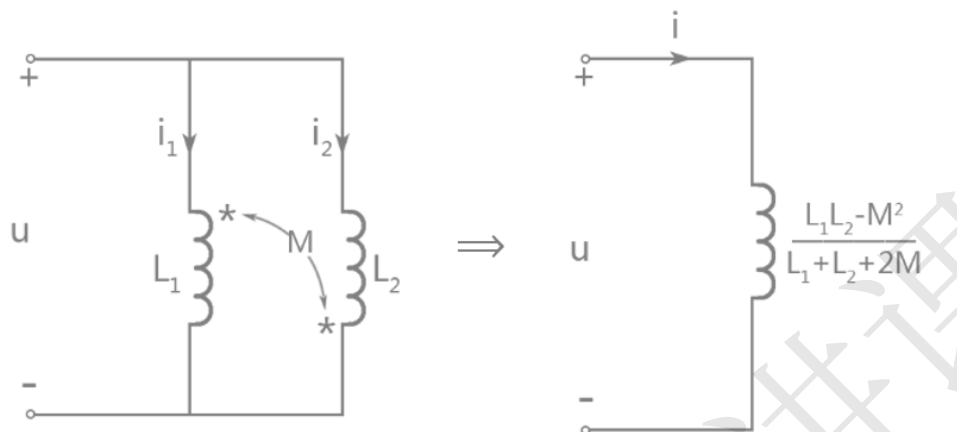
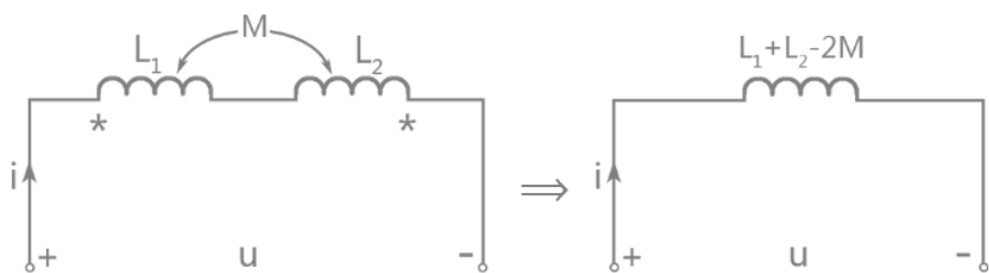
$$S_N = \sqrt{P^2 + Q^2} = \sqrt{46.19^2 + 11.52^2}$$

$$\bar{S} = P + jQ = 46.19 + 11.52j$$

电路第八课

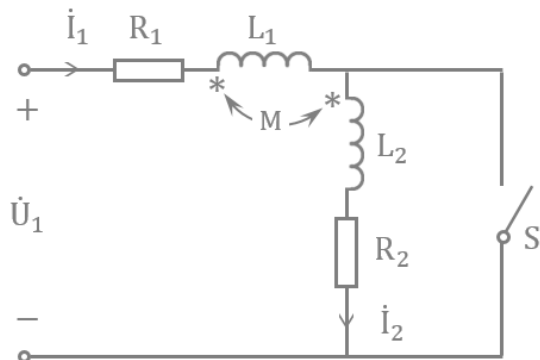
一、含有耦合电感的电路的等效



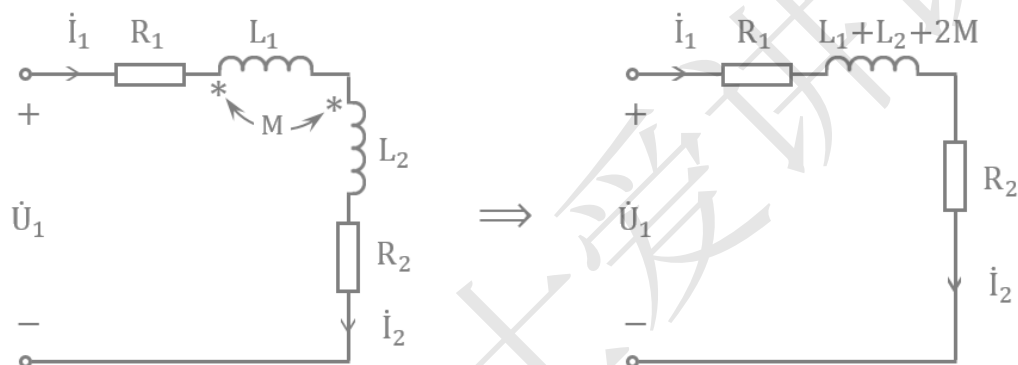


元件	电阻 R	电感 L	电容 C
Z 值	R	$j\omega L$	$\frac{-j}{\omega C}$

例 1: $R_1=R_2=1\Omega$, $\omega L_1=3\Omega$, $\omega L_2=2\Omega$, $\omega M=2\Omega$, $U_1=100V$, 求开关S打开与闭合时的电流 i_1 和 i_2



①开关 S 断开时

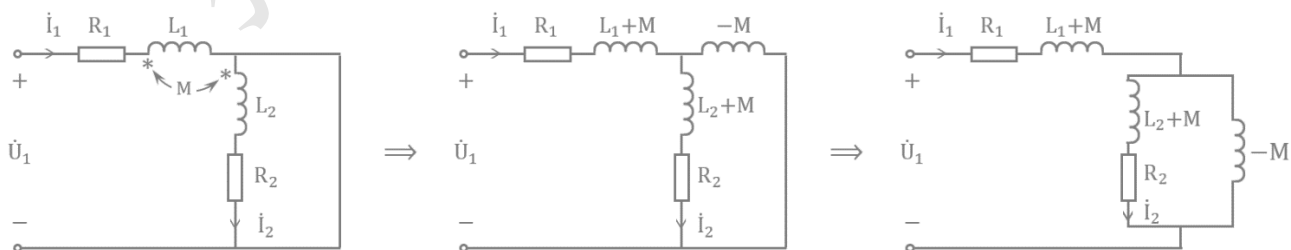


令 $\dot{U}_1 = 100 \angle 0^\circ \text{ V}$

$$Z_{\text{总}} = Z_{R_1} + Z_L + Z_{R_2} = 1 + j\omega(L_1 + L_2 + 2M) + 1 = 2 + 9j = 9.22 \angle 77.41^\circ \text{ A}$$

$$\dot{i}_1 = \dot{i}_2 = \frac{\dot{U}_1}{Z_{\text{总}}} = \frac{100 \angle 0^\circ}{9.22 \angle 77.41^\circ} = 10.85 \angle -77.41^\circ \text{ A}$$

②开关 S 闭合时



$$\begin{aligned}
 Z_{\text{总}} &= Z_{R_1} + Z_{L_1+M} + (Z_{L_2+M} + Z_{R_2}) // Z_{-M} \\
 &= R_1 + j\omega(L_1 + M) + [j\omega(L_2 + M) + R_2] // j\omega(-M) \\
 &= 1 + 5j + (1 + 4j) // (-2j) = 1 + 5j + \frac{(1+4j)(-2j)}{1+4j-2j} \\
 &= 1.8 + 1.4j \\
 &= 2.28 \angle 37.87^\circ
 \end{aligned}$$

$$\text{令 } \dot{U}_1 = 100 \angle 0^\circ \text{ V}$$

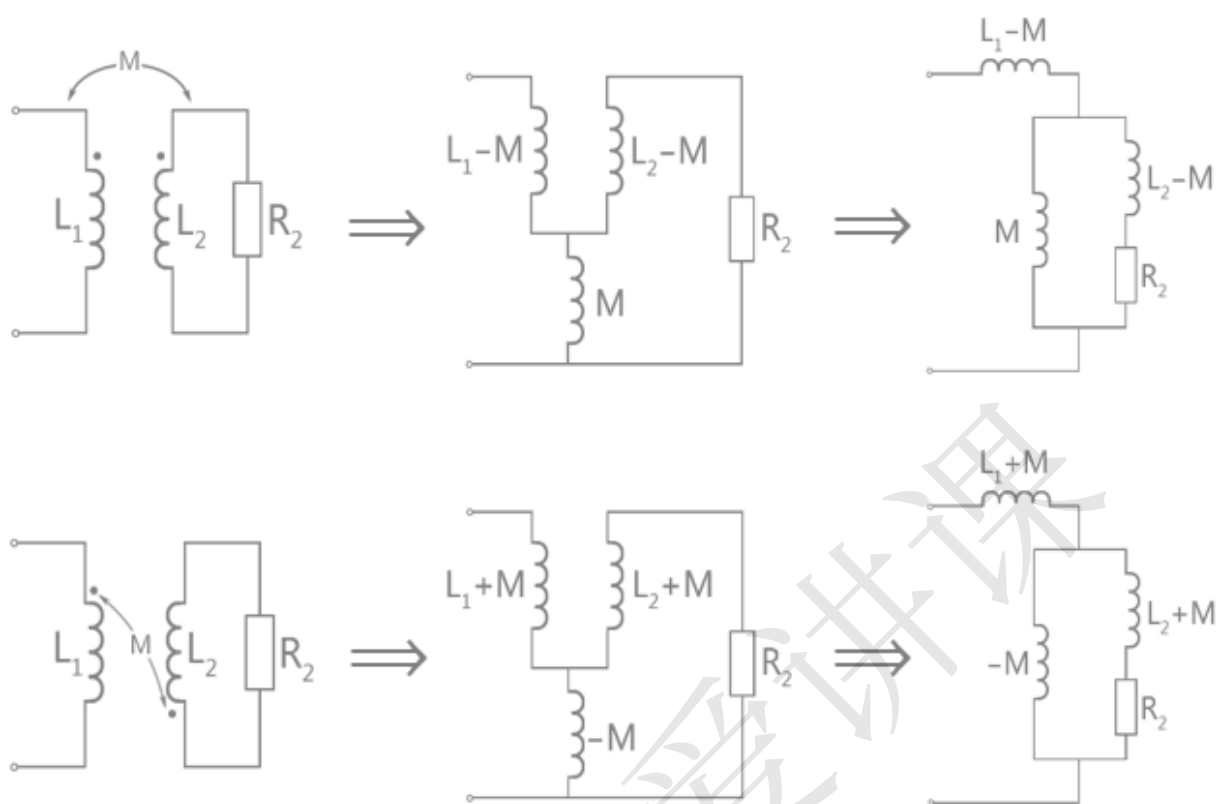
$$\dot{I}_1 = \frac{\dot{U}_1}{Z_{\text{总}}} = \frac{100 \angle 0^\circ}{2.28 \angle 37.87^\circ} = 43.85 \angle -37.87^\circ \text{ A}$$

$$Z_{2\text{总}} = Z_{L_2+M} + Z_{R_2} = j\omega(L_2 + M) + R_2 = 1 + 4j = 4.12 \angle 75.96^\circ$$

$$\begin{aligned}
 \text{则 } \dot{U}_2 &= \dot{U}_1 - \dot{U}_{R_1} - \dot{U}_{L_1+M} = 100 \angle 0^\circ - \dot{I}_1 \cdot Z_{R_1} - \dot{I}_1 \cdot Z_{L_1+M} = 100 \angle 0^\circ - 43.85 \angle -37.87^\circ \cdot (1) - 43.85 \angle -37.87^\circ \cdot (5j) \\
 &= 100 + 0j - (34.6 - 26.9j) \cdot (1) - (34.6 - 26.9j) \cdot (5j) = 161.59 \angle -115.35^\circ
 \end{aligned}$$

$$\dot{I}_2 = \frac{\dot{U}_2}{Z_{2\text{总}}} = \frac{161.6 \angle -115.35^\circ}{4.12 \angle 75.96^\circ} = 39.22 \angle -191.31^\circ \text{ A} = 39.22 \angle 168.69^\circ \text{ A}$$

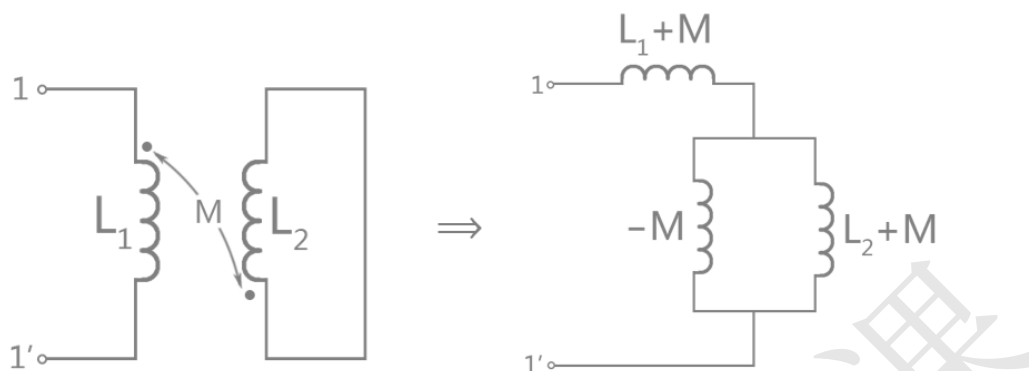
二、求空心变压器的参数



元件	电阻 R	电感 L	电容 C
Z 值	R	$j\omega L$	$\frac{-j}{\omega C}$

例 1：如图所示电路， $L_1=8\text{H}$ ， $L_2=2\text{H}$ ， $M=2\text{H}$ ，求从 1-1'端看进去的等效

电感 L



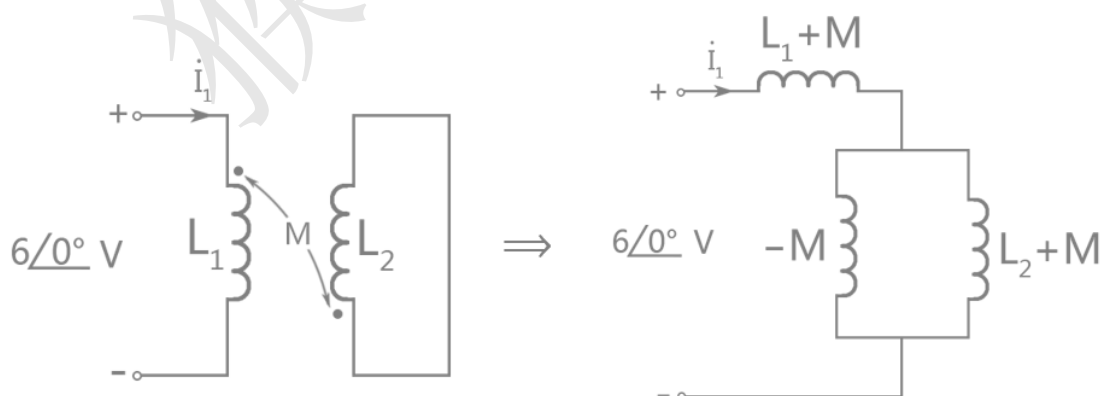
$$L = L_1 + M + (-M) // (L_2 + M)$$

$$= L_1 + M + \frac{(-M)(L_2 + M)}{-M + L_2 + M}$$

$$= 8 + 2 + \frac{(-2)(2+2)}{-2+2+2}$$

$$= 6\text{H}$$

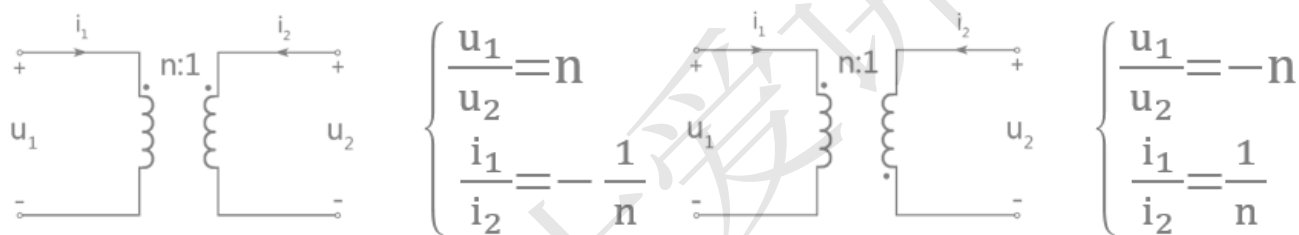
例 2：如图所示电路， $\omega L_1=8\Omega$ ， $\omega L_2=2\Omega$ ， $\omega M=2\Omega$ ，求电流 $\dot{I}_1$



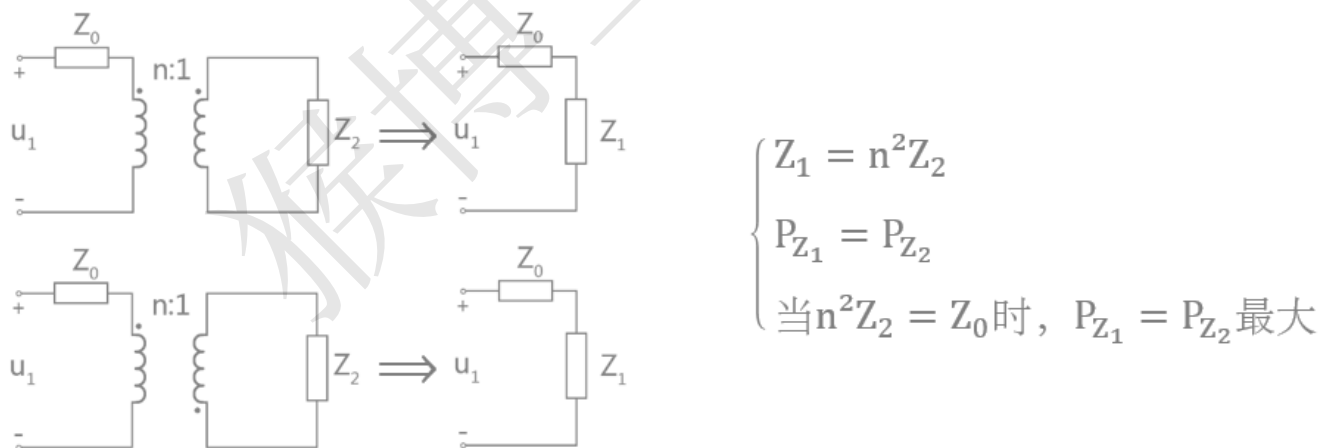
$$\begin{aligned}
 \dot{I}_1 &= \frac{\dot{U}}{Z_{\text{总}}} = \frac{6\angle 0^\circ}{j\omega(L_1+M) + j\omega(-M) // j\omega(L_2+M)} \\
 &= \frac{6\angle 0^\circ}{10j + (-2j) // 4j} \\
 &= \frac{6\angle 0^\circ}{6j} = \frac{6\angle 0^\circ}{6\angle 90^\circ} = 1\angle -90^\circ \text{ A}
 \end{aligned}$$

三、理想变压器

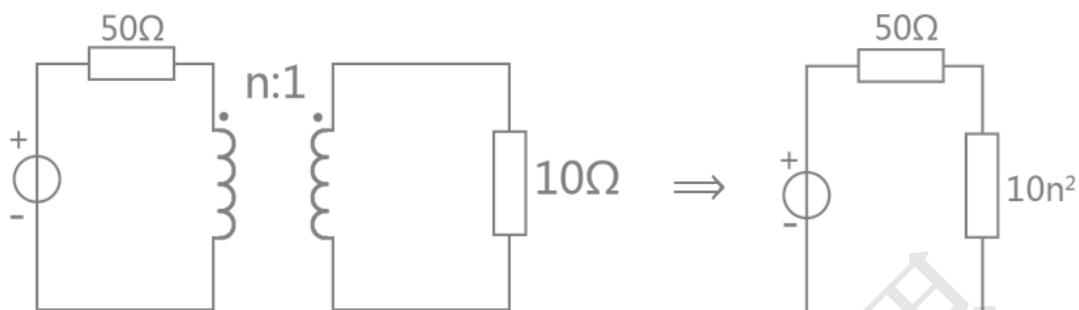
规律一：



规律二：



例 1: 如果 10Ω 电阻获得了最大功率, 请确定图示电路中理想变压器的变比 n



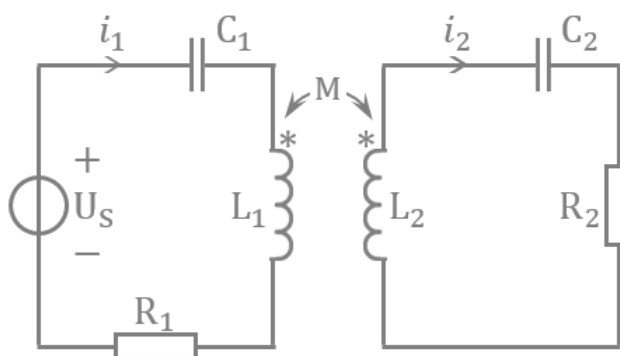
当 $n^2 \cdot 10 = 50$ 时, $P_{10n^2} = P_{10\Omega}$ 最大 $\Rightarrow n = \sqrt{5}$

四、求副线圈侧电阻的最大功率

元件	电阻 R	电感 L	电容 C
Z 值	R	$j\omega L$	$\frac{-j}{\omega C}$

例 1: 如图所示电路, $\dot{U}_S=10\angle 0^\circ \text{ V}$, $R_1=10\Omega$, $L_1=L_2=0.1\text{mH}$, $M=0.02\text{mH}$,

$C_1=C_2=0.01\mu\text{F}$, $\omega=10^6\text{rad/s}$, 求 R_2 为何值时获得最大功率, 并求出最大功率。



$$Z_{11}=Z_{C_1}+Z_{R_1}+Z_{L_1}$$

$$=\frac{-j}{\omega C_1}+R_1+j\omega L_1=\frac{-j}{10^6\times 0.01\times 10^{-6}}+10+j\times 10^6\times 0.1\times 10^{-3}=10$$

$$Z_{22}=Z_{C_2}+Z_{R_2}+Z_{L_2}$$

$$=\frac{-j}{\omega C_2}+R_2+j\omega L_2=\frac{-j}{10^6\times 0.01\times 10^{-6}}+R_2+j\times 10^6\times 0.1\times 10^{-3}=R_2$$

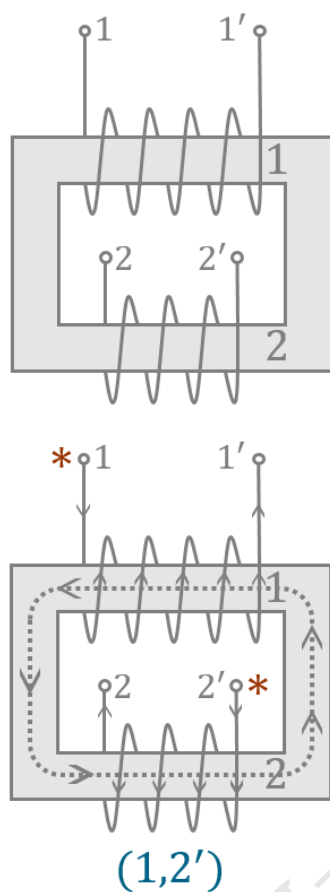
$$\frac{(\omega M)^2}{Z_{22}}=\frac{(10^6\times 0.02\times 10^{-3})^2}{R_2}=\frac{400}{R_2}$$

当 10 与 $\frac{400}{R_2}$ 共轭时 R_2 功率最大, 即 $R_2=40\Omega$ 时其功率最大,

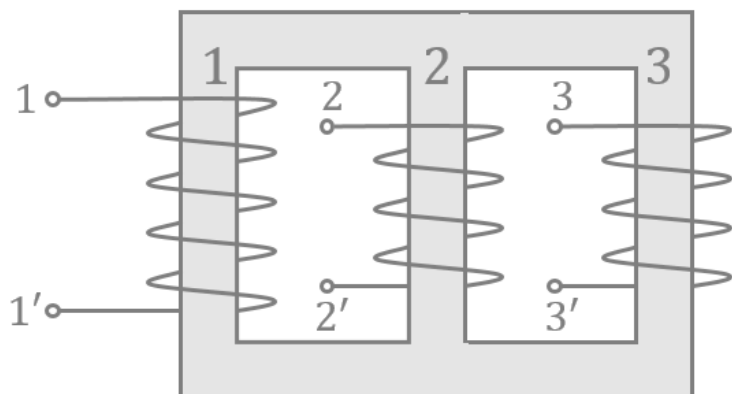
$$\text{最大功率 } P_{\max}=\frac{U_S^2}{4Z_{11\text{ 实数}}}=\frac{10^2}{4\times 10}=2.5\text{W}$$

五、确定同名端

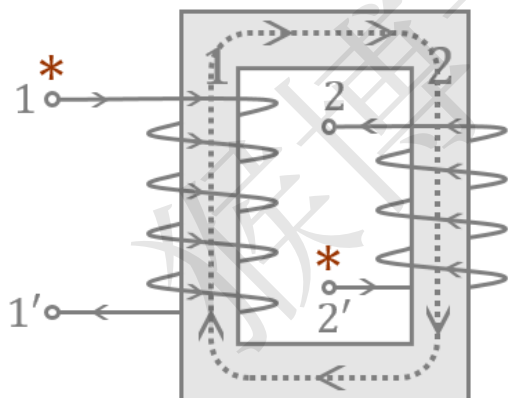
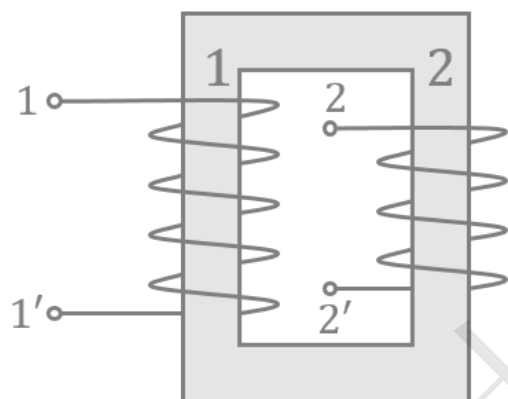
例 1：请确定图示线圈的同名端。



例 2：请确定图示线圈的同名端。

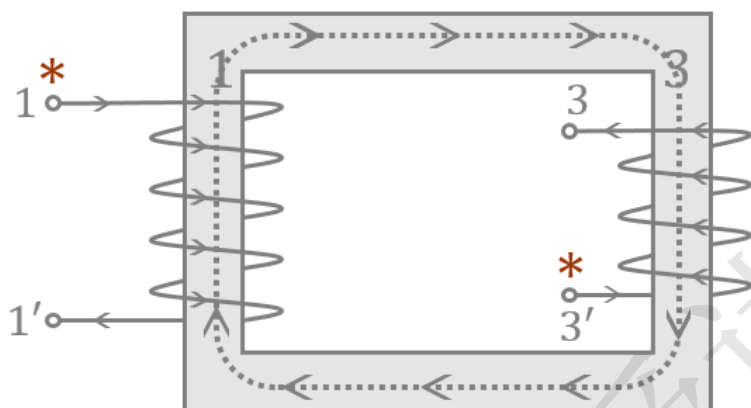
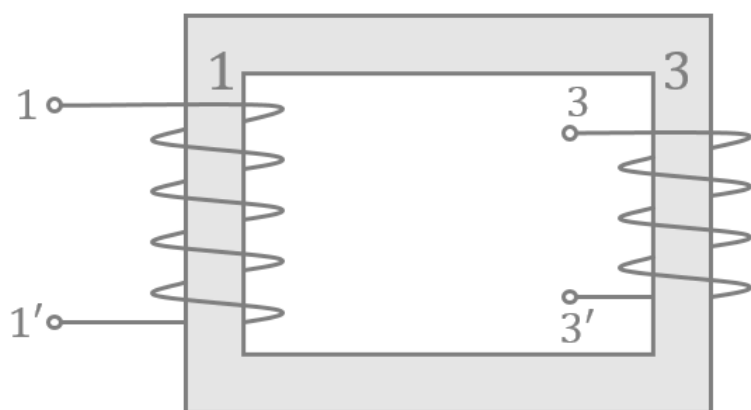


第一步：去掉线圈 3 及其所绕的铁芯，只判断线圈 1 和线圈 2

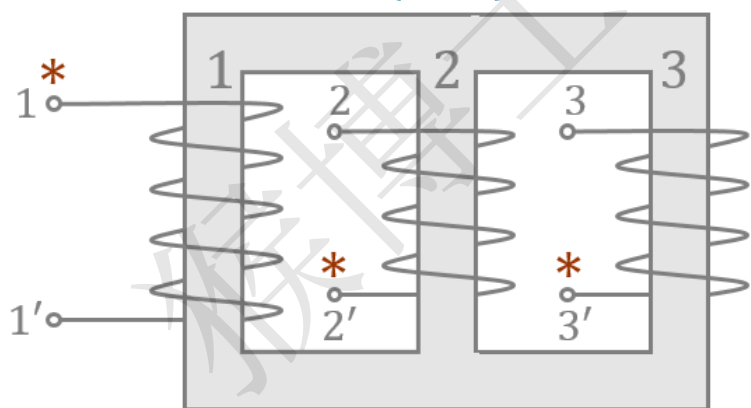


(1,2')

第二步：去掉线圈 2 及其所绕的铁芯，只判断线圈 1 和线圈 3



(1,3')



(1,2')

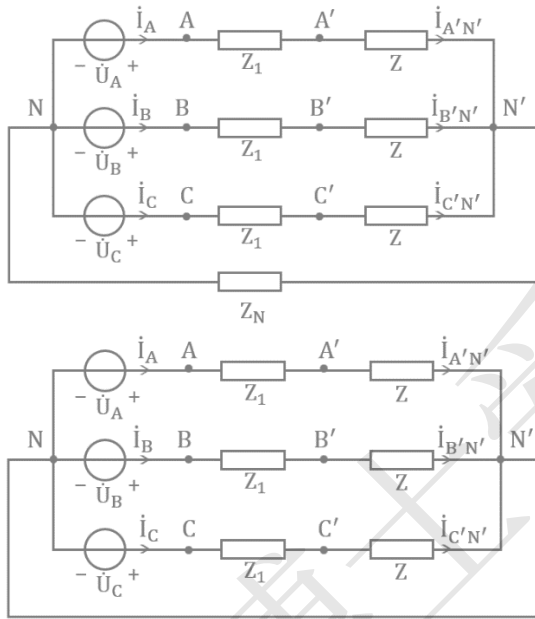
(1,3')

(注：线圈 2 和线圈 3 之间不用判断)

电路第九课

一、三相电路与 Y 型(星型)负载

例 1: 已知对称三相电路星型负载阻抗 $Z=(165+84j)\Omega$, 端线阻抗 $Z_1=(2+j)\Omega$, 中性线阻抗 $Z_N=(1+j)\Omega$, 线电压 $U_1=380V$ 。求负载端的相电流与线电压。



$$\text{线电压 } U_1=380V \Rightarrow U_{AB}=U_{BC}=U_{CA}=380V$$

$$\therefore U_A=U_B=U_C=\frac{380}{\sqrt{3}}=220V$$

$$\therefore \text{设 } \dot{U}_A=220\angle 0^\circ V$$

$$\dot{I}_{A'N'}=\frac{\dot{U}_A}{Z_1+Z}=\frac{220\angle 0^\circ}{(2+j)+(165+84j)}=1.174\angle -26.98^\circ A$$

$$\dot{I}_{B'N'}=\dot{I}_{A'N'}\angle -120^\circ=1.174\angle -146.98^\circ A$$

$$\dot{I}_{C'N'}=\dot{I}_{A'N'}\angle 120^\circ=1.174\angle 93.02^\circ A$$

$$\begin{aligned}\dot{U}_{A'B'}&=\sqrt{3}\dot{U}_{A'N'}\angle 30^\circ=\sqrt{3}\dot{I}_{A'N'}Z\angle 30^\circ \\ &=\sqrt{3}\times 1.174\angle -26.98^\circ\times(165+84j)\angle 30^\circ\end{aligned}$$

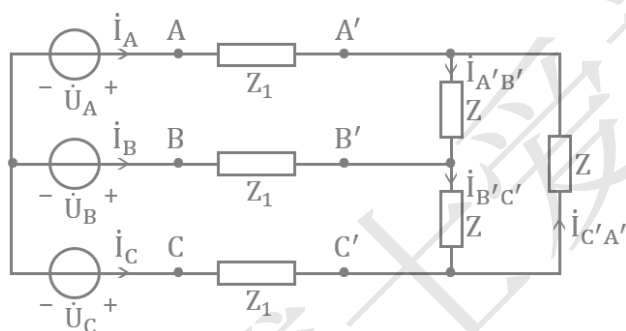
$$=376.5/\underline{30.05^\circ} \text{ V}$$

$$\dot{U}_{B'C'}=376.5/\underline{-89.95^\circ} \text{ V}$$

$$\dot{U}_{C'A'}=376.5/\underline{150.05^\circ} \text{ V}$$

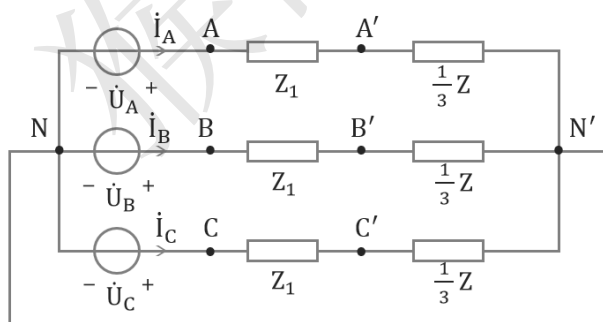
二、三相电路与 Δ 型(三角型)负载

例 1: 已知对称三相电路电源端电压 $U_1=220\text{V}$, 三角型负载阻抗 $Z=(495+252j)\Omega$, 端线阻抗 $Z_1=(2+j)\Omega$ 。求线电流、负载端相电流。



电源端电压 $U_1=220\text{V} \Rightarrow U_A=U_B=U_C=220\text{V}$

设 $\dot{U}_A=220/\underline{0^\circ}$



$$\dot{I}_A = \frac{\dot{U}_A}{Z_1 + \frac{1}{3}Z} = \frac{220\angle 0^\circ}{2+j + \frac{495+252j}{3}} = 1.174\angle -26.98^\circ$$

$$\dot{I}_{A'B'} = \frac{\sqrt{3}}{3} \dot{I}_A \angle 30^\circ = 0.678\angle 3.02^\circ$$

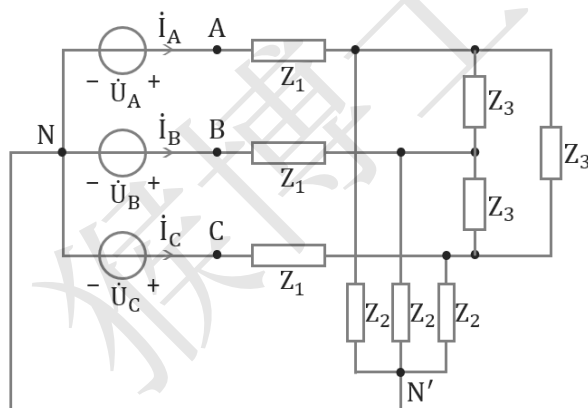
$$\dot{I}_B = \dot{I}_A \angle -120^\circ = 1.174\angle -146.98^\circ$$

$$\dot{I}_C = \dot{I}_A \angle 120^\circ = 1.174\angle 93.02^\circ$$

$$\dot{I}_{B'C'} = \dot{I}_{A'B'} \angle -120^\circ = 0.678\angle -116.98^\circ$$

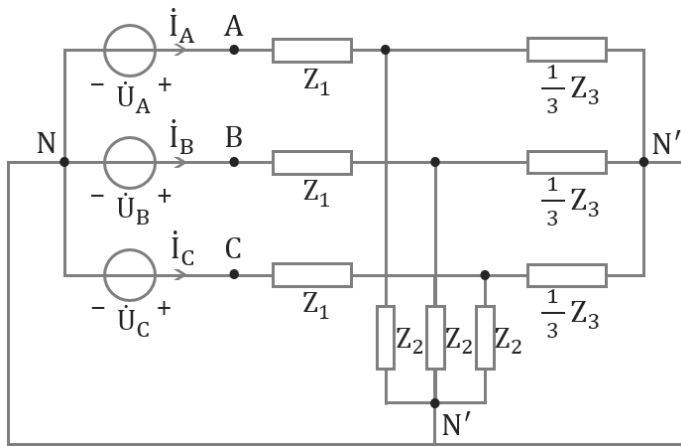
$$\dot{I}_{C'A'} = \dot{I}_{A'B'} \angle 120^\circ = 0.678\angle 123.02^\circ$$

例 2：已知对称三相电路电源端电压 $U_1=220V$ ，星型负载阻抗 $Z_2=(230+168j)\Omega$ ，三角型负载阻抗 $Z_3=(990+504j)\Omega$ ，端线阻抗 $Z_1=(2+j)\Omega$ 。求线电流。



电源端电压 $U_1=220V \Rightarrow U_A=U_B=U_C=220V$

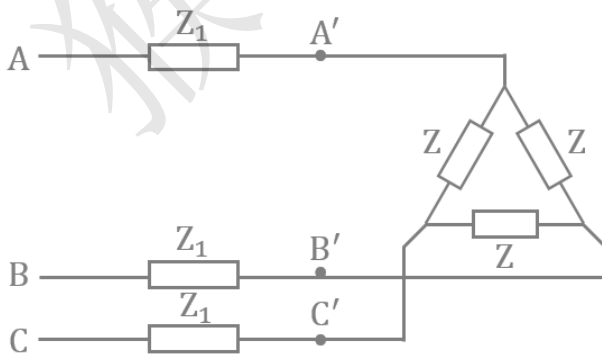
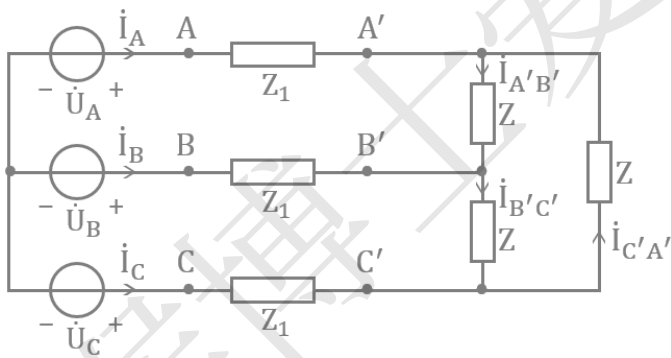
设 $\dot{U}_A=220\angle 0^\circ$



$$\dot{I}_A = \frac{\dot{U}_A}{Z_1 + Z_2 // \frac{1}{3}Z_3} = \frac{220/\underline{0^\circ}}{2 + j + (230 + 168j) // \frac{990 + 504j}{3}} = 1.174/\underline{-26.98^\circ} \text{ A}$$

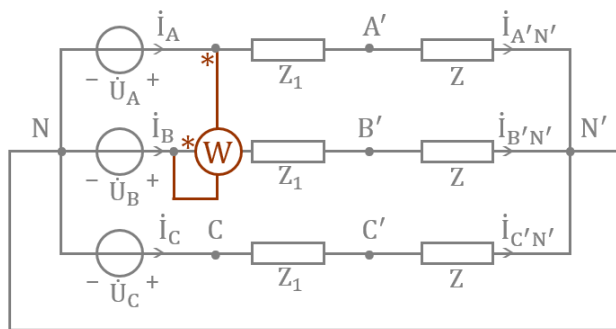
$$\dot{I}_B = \dot{I}_A / \underline{-120^\circ} = 1.174/\underline{-146.98^\circ} \text{ A}$$

$$\dot{I}_C = \dot{I}_A / \underline{120^\circ} = 1.174/\underline{93.02^\circ} \text{ A}$$



三、求三相电路里的功率表读数 P

例 1: 已知对称三相电路星型负载阻抗 $Z=(165+84j)\Omega$, 端线阻抗 $Z_1=(2+j)\Omega$, 中性线阻抗 $Z_N=(1+j)\Omega$, 线电压 $U_1=380V$ 。求负载端的相电流与线电压。



$$\text{线电压 } U_1=380V \Rightarrow U_{AB}=U_{BC}=U_{CA}=380V$$

$$\therefore U_A=U_B=U_C=\frac{380}{\sqrt{3}}=220V$$

$$\therefore \text{设 } \dot{U}_A=220\angle 0^\circ V$$

$$\dot{I}_{A'N'}=\frac{\dot{U}_A}{Z_1+Z}=\frac{220\angle 0^\circ}{(2+j)+(165+84j)}=1.174\angle -26.98^\circ A$$

$$\dot{I}_{B'N'}=\dot{I}_{A'N'}\angle -120^\circ=1.174\angle -146.98^\circ A$$

$$\dot{I}_{C'N'}=\dot{I}_{A'N'}\angle 120^\circ=1.174\angle 93.02^\circ A$$

$$\dot{U}_{AB}=\sqrt{3}\dot{U}_A\angle 30^\circ=380\angle 30^\circ V$$

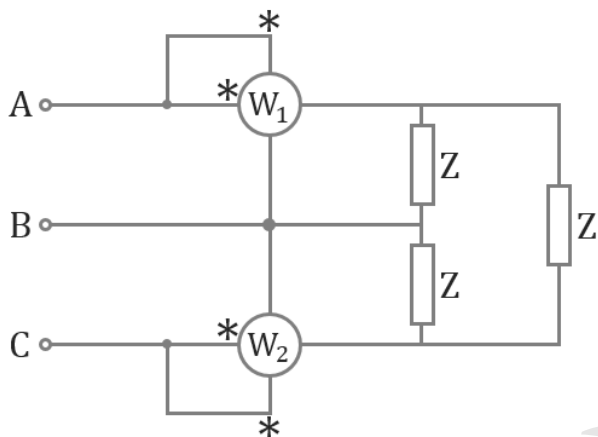
$$\dot{I}_B=\dot{I}_{B'N'}=1.174\angle -146.98^\circ A$$

$$P=U_{AB}I_B\cos(\varphi_{AB}-\varphi_B)$$

$$=380\times 1.174\times \cos[30^\circ-(-146.98^\circ)]$$

四、电路里接两表且读数已知，求有功/无功/视在/复功率

例 1: 三角型对称三相电路如图所示, $U_{AB}=380V$, 功率表 $W_1=782W$, $W_2=1976.44W$ 。求电路的复功率 $\bar{S}$ 。



① 设 $\dot{U}_A = U \angle 0^\circ$ $\dot{I}_A = I \angle -\varphi$

$$U_{AB} = 380V \Rightarrow U_A = \frac{380V}{\sqrt{3}} = 220V$$

$$\therefore \dot{U}_A = 220 \angle 0^\circ V$$

② $\dot{U}_{AB} = \sqrt{3} \dot{U}_A \angle 30^\circ = 380 \angle 30^\circ V$

$$\dot{U}_{BC} = \dot{U}_{AB} \angle -120^\circ = 380 \angle -90^\circ V$$

$$\dot{U}_{CA} = \dot{U}_{AB} \angle 120^\circ = 380 \angle 150^\circ V$$

$$\dot{I}_B = \dot{I}_A \angle -120^\circ = I \angle -120^\circ - \varphi$$

$$\dot{I}_C = \dot{I}_A \angle 120^\circ = I \angle 120^\circ - \varphi$$

③ $P_1 = U_{AB} I_A \cos(\varphi_{AB} - \varphi_A)$

$$= 380 \cdot I \cdot \cos[30^\circ - (-\varphi)] = 380 I \cos(\varphi + 30^\circ)$$

$$P_2 = U_{CB} I_C \cos(\varphi_{CB} - \varphi_C)$$

$$= 380 \cdot I \cdot \cos[90^\circ - (120^\circ - \varphi)] = 380 I \cos(\varphi - 30^\circ)$$

$$\frac{P_1}{P_2} = \frac{380 I \cos(\varphi + 30^\circ)}{380 I \cos(\varphi - 30^\circ)} = \frac{\cos(\varphi + 30^\circ)}{\cos(\varphi - 30^\circ)} = \frac{782}{1976.44} \Rightarrow \varphi = 36.87^\circ \quad \tan \varphi = 0.75$$

④ $P = P_1 + P_2 = 782 + 1976.44 = 2758.44 W$

$$Q = P \tan \varphi = 2758.44 \times 0.75 = 2068.83 \text{ Var}$$

$$\bar{S} = P + jQ = (2758.44 + j2068.83) \text{ VA}$$